Finite Element Methods

MATH 6660 — Fall 2026

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lecture transcription

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Chapter 1

Lectures

1.1 Why FEM, and the two-point BVP

Lecture 1 — Monday, August 24, 2026 Math 6660

Context

Where this course sits relative to the rest of the numerical sequence:

Prerequisites	The discretisation courses
5610: interpolation, integration, ODEs	**6660**: FEMs (this course)
5620: numerical linear algebra	6670: FVMs
	5170: FDMs

Why FEM?

(1) Theoretical. The original problem is continuous.

- **FDM**: approximate the solution at a discrete number of points.
- **FEM**: keep the original continuous problem (the PDE), but replace (approximate) the *function spaces*.

All the theory of PDEs carries over to FEM.

(2) Practical reasons.

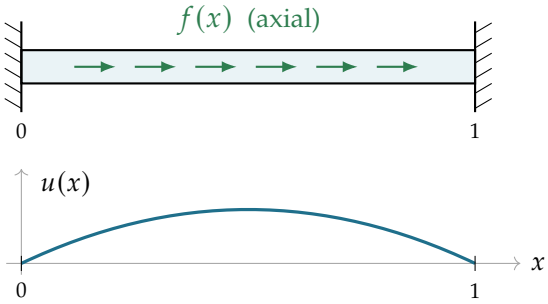
- Can be formulated on an unstructured grid.
- Lots of flexibility in the choice of function spaces.

(3) Note. Much of the theory of FEMs is based on *elliptic* PDE theory. Parabolic PDEs are OK. Hyperbolic PDEs are more challenging.

The two-point BVP

The 1D version of Poisson's equation. On the fixed domain $0 < x < 1$:

$$-\big(a(x)\,u'\big)' = f(x), \qquad u(0) = u(1) = 0. \tag{1.1}$$



u := "displacement"
f := "external load" given
a = "modulus of elasticity" given

Assume:

$$0 < a_1 \le a(x) \le a_2 < \infty \qquad \forall\, x \in [0,1]. \tag{1.2}$$

The operator is self-adjoint

Note that

$$\mathcal{L}(u) = -(au')' \qquad \text{with} \qquad u(0) = u(1) = 0 \tag{1.3}$$

is a linear differential operator, and $\mathcal{L}$ is *self-adjoint*.

To see it, integrate by parts twice:

$$\begin{aligned}\int_0^1 v\,\mathcal{L}(u)\,\mathrm{d}x &= -\int_0^1 v\,(au')'\,\mathrm{d}x \\ &= \int_0^1 v'\,a\,u'\,\mathrm{d}x - \Big[v\,a\,u'\Big]_{x=0}^{x=1}.\end{aligned} \tag{1.4}$$

Assume $u(0) = u(1) = 0$ and $v(0) = v(1) = 0$; the boundary term drops:

$$\begin{aligned}&= \int_0^1 v'\,a\,u'\,\mathrm{d}x \\ &= -\int_0^1 u\,(av')'\,\mathrm{d}x + \Big[u\,a\,v'\Big]_{x=0}^{x=1} \\ &= -\int_0^1 u\,(av')'\,\mathrm{d}x = \int_0^1 u\,\mathcal{L}(v)\,\mathrm{d}x.\end{aligned} \tag{1.5}$$

$$\boxed{\int_0^1 v\,\mathcal{L}(u)\,\mathrm{d}x = \int_0^1 u\,\mathcal{L}(v)\,\mathrm{d}x} \qquad v = \text{test function} \tag{1.6}$$

This is what it means to be self-adjoint.

Note. A self-adjoint differential operator $\mathcal{L}$ is analogous to a symmetric matrix.

Invertibility and regularity

Note. Consider

$$\mathcal{L}(w) = 0, \qquad w(0) = w(1) = 0.$$

The only solution that satisfies $\mathcal{L}$ and the BCs is $w = 0$. Therefore $\mathcal{L}$ is "invertible": for given a and f there exists a unique u.

Note (regularity). If $0 < a < \infty$ and f are sufficiently smooth — specifically, if

$$a \in C^1((0,1)) \quad \text{and} \quad f \in C^0((0,1)),$$

then $u \in C^2((0,1))$.

Variational formulation

Multiply our BVP by a test function and integrate over the domain $[0,1]$:

$$-\int_0^1 (au')'\, v \,\mathrm{d}x = \int_0^1 \underbrace{f}_{\text{force}}\, v \,\mathrm{d}x. \tag{1.7}$$

Then integrate by parts:

$$\int_0^1 a\, u'v' \,\mathrm{d}x - \Big[a\, u'v\Big]_0^1 = \int_0^1 f v \,\mathrm{d}x. \tag{1.8}$$

Assume v satisfies the BCs:

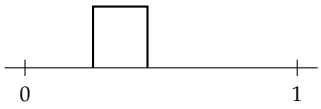
$$v(0) = v(1) = 0.$$

In particular, define the solution space

$$V = \left\{ v \;:\; \int_0^1 v^2 \,\mathrm{d}x < \infty, \quad \int_0^1 \big[v'\big]^2 \,\mathrm{d}x < \infty, \quad v(0) = v(1) = 0 \right\}. \tag{1.9}$$

Note. Functions in V are continuous.

(1) $\displaystyle\int_0^1 v^2 \,\mathrm{d}x < \infty$ *admits* discontinuous functions:



(2) $\displaystyle\int_0^1 \big[v'\big]^2 \,\mathrm{d}x$ does *not* allow discontinuities in V.

Summary: what the strong and weak forms each demand

	Strong form ($u \in C^2((0,1))$)	**Weak form** ($u \in V = H_0^1$)
demands on u	u, u', u'' all continuous	u continuous, $\int (u')^2 < \infty$
u' may jump?	no ×	**yes** ✓
u'' required?	yes	never appears
equation holds	at every point	in an integral (averaged) sense

What the weak form buys.

	strong	**weak**
material interface (a jumps)	no solution ×	fine ✓
point load, re-entrant corner	no solution ×	fine ✓
hat functions	illegal ×	legal ✓ (FEM exists only here)
element must compute	u''	only u' (the $\boldsymbol{B}$ matrix)
natural BC (traction)	imposed by hand	free — appears in the load vector
existence/uniqueness	no theory	Lax–Milgram (H^1 is Hilbert)
sparsity	—	small-support kinked basis ⇒ banded $\boldsymbol{K}$

What it costs.

- The PDE cannot be checked at a point — only weighted averages of it.
- Strain and stress come out *discontinuous* across element boundaries; contour plots need smoothing before presentation.
- Derivatives are one order less accurate than the displacement, and are best sampled at Gauss points rather than nodes.
- Conceptual overhead: Sobolev spaces and weak derivatives, i.e. the first two lectures of this course.

1.2 Weak solutions: the three claims

Lecture 2 Math 6660 — continues in the same OneNote export, from p. 7

Setup, restated

Two-point BVP. On the fixed domain $0 < x < 1$:

$$-(au')' = f, \qquad u(0) = u(1) = 0, \qquad \text{find } u \in C^2((0,1)). \tag{1.10}$$

Variational formulation.

$$V = \left\{ v \ : \ \int_0^1 v^2 \,\mathrm{d}x < \infty, \quad \int_0^1 (v')^2 \,\mathrm{d}x < \infty, \quad v(0) = v(1) = 0 \right\} \qquad \text{(defining a space)} \tag{1.11}$$

Note. Functions in V are continuous $\big(v \in C^0((0,1))\big)$.

Example of a discontinuity

Let

$$H(x) = \begin{cases} 1, & \frac{1}{4} < x < \frac{3}{4}, \\ 0, & \text{otherwise}, \end{cases} \tag{1.12}$$

which is not continuous: $H \notin C^0((0,1))$, but $H \in C^{-1}((0,1))$ — it has a finite jump.

So

$$\int_0^1 H^2 \,\mathrm{d}x = \int_{1/4}^{3/4} \mathrm{d}x = \tfrac{1}{2} \ < \ \infty. \tag{1.13}$$

The first condition in (1.11) is satisfied. Now the derivative:

$$H'(x) = \delta\big(x - \tfrac{1}{4}\big) - \delta\big(x - \tfrac{3}{4}\big) \ = \ \delta_{1/4} - \delta_{3/4}, \tag{1.14}$$

writing $\delta_{1/4} \equiv \delta(x - \frac{1}{4})$ and $\delta_{3/4} \equiv \delta(x - \frac{3}{4})$ — the same objects, shorter names. Each jump contributes one delta, sited at the jump, with coefficient equal to the *signed* jump size:

at	H **goes**	**size**	**contributes**
$x = \frac{1}{4}$	$0 \to 1$	$+1$	$+\,\delta_{1/4}$
$x = \frac{3}{4}$	$1 \to 0$	-1	$-\,\delta_{3/4}$

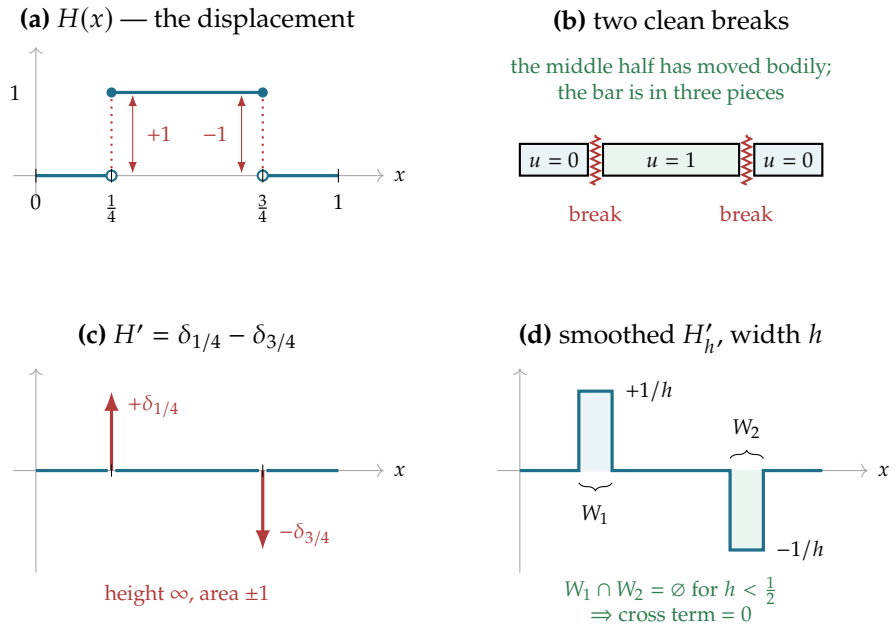


Figure 1.1: The step H: (a) the displacement field, jumping +1 then −1; (b) the same thing as a bar — three pieces, two breaks; (c) its derivative, a pair of deltas; (d) the smoothed version used for the honest calculation.

Recall (δ properties). For $\delta(x - x_0)$:

$$\int_{x_0-\varepsilon_1}^{x_0+\varepsilon_2} \delta(x - x_0)\,dx = 1 \qquad \text{for any } \varepsilon_1, \varepsilon_2 > 0, \tag{1.15}$$

$$\int_{x_0-\varepsilon_1}^{x_0+\varepsilon_2} f(x)\,\delta(x - x_0)\,dx = f(x_0), \qquad f \text{ a continuous function.} \tag{1.16}$$

Now expand the square — three terms, and *all three* must be accounted for:

$$\begin{aligned}\int_0^1 (H')^2\,dx &= \int_0^1 \left(\delta_{1/4} - \delta_{3/4}\right)^2 dx \\ &= \int_0^1 \underbrace{\delta_{1/4}^2}_{\text{(i)}}\,dx \;-\; 2\int_0^1 \underbrace{\delta_{1/4}\,\delta_{3/4}}_{\text{(ii)}}\,dx \;+\; \int_0^1 \underbrace{\delta_{3/4}^2}_{\text{(iii)}}\,dx.\end{aligned} \tag{1.17}$$

An arrow is then drawn from term (i) down to the assertion

$$\int_0^1 \delta\left(x - \tfrac{1}{4}\right)\delta\left(x - \tfrac{1}{4}\right)\,dx = +\infty, \tag{1.18}$$

and the conclusion $H \notin V$ follows immediately.

The honest computation. Replace each jump by a ramp of width h. Then H'_h is an ordinary piecewise-constant *function*:

$$H'_h(x) = \begin{cases} +1/h, & x \in W_1 = \left(\frac{1}{4} - \frac{h}{2},\ \frac{1}{4} + \frac{h}{2}\right), \\ -1/h, & x \in W_2 = \left(\frac{3}{4} - \frac{h}{2},\ \frac{3}{4} + \frac{h}{2}\right), \\ 0, & \text{elsewhere,} \end{cases} \qquad \text{and for } h < \tfrac{1}{2},\ W_1 \cap W_2 = \varnothing. \tag{1.19}$$

Now every term of (1.17) can be evaluated with no distributions at all:

	term	smoothed integrand	integral
(i)	$\delta_{1/4}^2$	$(1/h)^2$ on W_1, else 0	$\frac{1}{h^2}\cdot h = \frac{1}{h}$
(ii)	$-2\,\delta_{1/4}\delta_{3/4}$	0 everywhere ($W_1 \cap W_2 = \varnothing$)	0
(iii)	$\delta_{3/4}^2$	$(-1/h)^2$ on W_2, else 0	$\frac{1}{h^2}\cdot h = \frac{1}{h}$
	total		$\frac{1}{h} + 0 + \frac{1}{h} = \frac{2}{h} \longrightarrow \infty$

Note (ii): the cross term vanishes for a *rigorous* reason, not a hand-wave — the two windows are disjoint, so wherever one factor is nonzero the other is zero. **Deltas at different places never interact.** And (iii) contributes $+1/h$, not $-1/h$: the sign is destroyed by the square, so the two spikes *add*.

Recall (the plain integral passes — as always). Run (1.19) through the *unsquared* test as a control:

$$\int_0^1 H_h' \,\mathrm{d}x = \underbrace{\left(+\tfrac{1}{h}\right) h}_{W_1} + \underbrace{\left(-\tfrac{1}{h}\right) h}_{W_2} = 1 - 1 = 0 \;=\; H(1) - H(0) \quad \checkmark \text{ (FTC)}.$$

Finite for every h, and independent of h. So the plain condition would *admit* this broken bar. Only the square rejects it.

Therefore

$$\boxed{H(x) \notin V} \qquad \text{because } H' \text{ is not square integrable.} \tag{1.20}$$

The variational problem

$$\text{Find } u \in V \text{ such that } \int_0^1 a\, u'v' \,\mathrm{d}x = \int_0^1 f v \,\mathrm{d}x \qquad \text{for all } v \in V. \tag{1.21}$$

Note. u is the *trial* function; v are the *test* functions. In this case u and v come from the same function space, V.

Q: How are (2-pt BVP) and (variational problem) related?

- BVP requires $u \in C^2((0,1))$.
- Variational problem only needs $u \in V$.

Claim 1: uniqueness

Proposition 1.2.1. *If the variational problem has a solution, it is unique.*

Proof (by contradiction). Assume there exist 2 different solutions, u and U:

$$\int_0^1 a u'v' \,\mathrm{d}x = \int_0^1 f v \,\mathrm{d}x \qquad \Big| \qquad \int_0^1 a U'v' \,\mathrm{d}x = \int_0^1 f v \,\mathrm{d}x.$$

Then subtract:

$$\Rightarrow \quad \int_0^1 a\,(u-U)'\,v' \,\mathrm{d}x = 0. \tag{1.22}$$

Take $v = u - U \in V$. Now

$$\int_0^1 a\Big((u-U)'\Big)^2 \,\mathrm{d}x = 0 \quad \Longrightarrow \quad (u-U)' = 0 \text{ on } (0,1) \quad \Longrightarrow \quad u = U + c. \tag{1.23}$$

But by the BCs, $c = 0$, so $u = U$. □

Remark 1.2.2. The lecturer labels this "by contradiction," but no contradiction is actually reached — the argument derives $u = U$ directly. It is a direct uniqueness proof.

Claim 2: BVP ⇒ variational

Proposition 1.2.3. *A solution to the BVP is a solution to the variational problem.*

Proof. The BVP has a unique solution $u \in C^2((0,1))$ with $u(0) = u(1) = 0$. If $u \in C^2((0,1))$ and satisfies the BCs, then $u \in V$. The BVP and the variational problem are then equivalent via integration by parts. □

Claim 3: variational ⇒ BVP

Proposition 1.2.4. *If $f \in C^0((0,1))$, $a \in C^1((0,1))$, and $0 < a_1 \le a(x) \le a_2 < \infty$ for all $x \in (0,1)$, then a solution to the variational problem is a solution to the BVP.*

Proof. Assume for the moment that $u \in C^2((0,1))$. From (1.21),

$$\int_0^1 \big(au'v' - fv\big)\,\mathrm{d}x = 0. \tag{1.24}$$

Reverse integration by parts:

$$\int_0^1 \underbrace{\Big[(au')' + f\Big]}_{S(x)} v\,\mathrm{d}x = 0. \tag{1.25}$$

Goal: show $S(x) = 0$ for all $x \in (0,1)$.

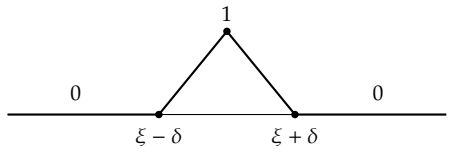
(a) Assume, for contradiction, that

$$S(\xi) = \Big[(au')' + f\Big]\Big|_{x=\xi} > 0 \qquad \text{at some point } \xi \in (0,1).$$

Note that $S(x)$ is continuous in x. By continuity, $S(x) > 0$ for $x \in [\xi - \delta,\ \xi + \delta]$ for some $\delta > 0$.

Choose v to be the following "hat" function:

$v =$



But then

$$\int_0^1 S(x)\,v(x)\,\mathrm{d}x > 0,$$

which is a contradiction with (1.25). Therefore $S(x)$ cannot be positive in $(0,1)$.

(b) Redo this argument with $S(\xi) < 0 \Rightarrow S(x)$ cannot be negative in $(0,1)$.

Hence $S \equiv 0$, which is (1.10). □

When can we expect $u \in C^2((0,1))$?

$$(\star) \qquad f \in C^0((0,1)), \quad a \in C^1((0,1)), \quad \text{and} \quad 0 < a_1 \le a(x) \le a_2 < \infty \quad \forall\, x \in (0,1). \tag{1.26}$$

Note.

① If $(\star)$ is satisfied: u is a *strong* solution of the 2-pt BVP, and

$$\text{(2-pt BVP)} \ \equiv\ \text{(variational problem)}.$$

② If $(\star)$ is *not* satisfied, the 2-pt BVP will not have a solution, but the variational problem may still have one. Then u (if it exists) is a *weak* solution of the 2-pt BVP.

1.3 Energy minimisation and the first finite element method

Setup, restated

2-pt BVP.

$$-(au')' = f \ \text{ on } (0,1), \qquad u(0) = u(1) = 0, \tag{1.27}$$

with $f \in C^0((0,1))$, $a \in C^1((0,1))$, and $0 < a_1 \le a(x) \le a_2 < \infty$ for all $x \in [0,1]$.

Variational problem. Find $u \in V$ such that

$$\int_0^1 a\, u'v' \,\mathrm{d}x = \int_0^1 f v \,\mathrm{d}x \qquad \forall\, v \in V, \tag{1.28}$$

where $V = \left\{v : \int v^2 \,\mathrm{d}x < \infty,\ \int (v')^2 \,\mathrm{d}x < \infty,\ v(0) = v(1) = 0\right\}$.

The "energy" minimisation problem

Define the following energy functional:

$$F(w) \ = \ \underbrace{\frac{1}{2}\int_0^1 a\,(w')^2 \,\mathrm{d}x}_{\text{internal energy}} \ - \ \underbrace{\int_0^1 f\, w \,\mathrm{d}x}_{\text{potential energy}} \qquad w \text{ is a function.} \tag{1.29}$$

Min problem. Find $u \in V$ such that

$$F(u) \le F(w) \qquad \forall\, w \in V. \tag{1.30}$$

Claim: the Min Problem and the variational problem are equivalent

Theorem 1.3.1. *The minimisation problem* (1.30) *and the variational problem* (1.28) *are equivalent.*

① **(variational problem) ⇒ (Min Prob).** Assume $u \in V$ satisfies $\int_0^1 au'v' \,\mathrm{d}x = \int_0^1 fv \,\mathrm{d}x$ for all $v \in V$. Let $w \in V$, and set $w = u + v$. Then

$$\begin{aligned} F(w) &= F(u+v) \\ &= \frac{1}{2}\int_0^1 a\,(u'+v')^2 \,\mathrm{d}x - \int_0^1 fu \,\mathrm{d}x - \int_0^1 fv \,\mathrm{d}x \\ &= \frac{1}{2}\int_0^1 a(u')^2 \,\mathrm{d}x + \underbrace{\int_0^1 au'v' \,\mathrm{d}x}_{\text{(A)}} + \frac{1}{2}\int_0^1 a(v')^2 \,\mathrm{d}x - \int_0^1 fu \,\mathrm{d}x - \underbrace{\int_0^1 fv \,\mathrm{d}x}_{\text{(B)}}. \end{aligned} \tag{1.31}$$

Terms (A) and (B) cancel, since (A) − (B) = 0 is exactly the variational problem. What remains is

$$\boxed{F(w) = F(u) + \frac{1}{2}\int_0^1 a\,(v')^2 \,\mathrm{d}x} \tag{1.32}$$

and since $a \ge a_1 > 0$, the last term is ≥ 0. Therefore

$$F(w) \ \ge \ F(u) \qquad \forall\, w \in V. \tag{1.33}$$

Recall (this is just completing the square). In finite dimensions with $\boldsymbol{K}$ symmetric positive definite, $F(\boldsymbol{w}) = \frac{1}{2}\boldsymbol{w}^T\boldsymbol{K}\boldsymbol{w} - \boldsymbol{w}^T\boldsymbol{F}$. Putting $\boldsymbol{w} = \boldsymbol{u} + \boldsymbol{v}$ with $\boldsymbol{K}\boldsymbol{u} = \boldsymbol{F}$ gives

$$F(\boldsymbol{u}+\boldsymbol{v}) = F(\boldsymbol{u}) + \boldsymbol{v}^T(\boldsymbol{K}\boldsymbol{u} - \boldsymbol{F}) + \tfrac{1}{2}\boldsymbol{v}^T\boldsymbol{K}\boldsymbol{v} = F(\boldsymbol{u}) + \tfrac{1}{2}\boldsymbol{v}^T\boldsymbol{K}\boldsymbol{v},$$

which is (1.32) term for term. Solving $\boldsymbol{K}\boldsymbol{u} = \boldsymbol{F}$ and minimising F are the same task — exactly what Theorem 1.3.1 says in the continuous setting.

② **(Min Prob) ⇒ (variational problem).** Assume $u \in V$ satisfies $F(u) \le F(w)$ for all $w \in V$. Let

$$g(\varepsilon) = F(u + \varepsilon v), \qquad \text{where } v \in V \text{ is arbitrary but fixed.} \tag{1.34}$$

Expanding,

$$\begin{aligned} g(\varepsilon) &= \frac{1}{2}\int_0^1 a\,(u' + \varepsilon v')^2\,\mathrm{d}x - \int_0^1 f\,(u+\varepsilon v)\,\mathrm{d}x \\ &= \frac{1}{2}\int_0^1 a(u')^2\,\mathrm{d}x + \varepsilon\int_0^1 au'v'\,\mathrm{d}x + \frac{1}{2}\varepsilon^2\int_0^1 a(v')^2\,\mathrm{d}x - \int_0^1 fu\,\mathrm{d}x - \varepsilon\int_0^1 fv\,\mathrm{d}x. \end{aligned} \tag{1.35}$$

$g(\varepsilon)$ is a quadratic polynomial in ε with a positive quadratic term. The unique global minimiser happens at $\varepsilon = 0$, so $g'(0) = 0$. Since

$$g'(\varepsilon) = \int_0^1 a\,u'v'\,\mathrm{d}x - \int_0^1 fv\,\mathrm{d}x + \varepsilon\int_0^1 a\,(v')^2\,\mathrm{d}x, \tag{1.36}$$

setting $\varepsilon = 0$ gives

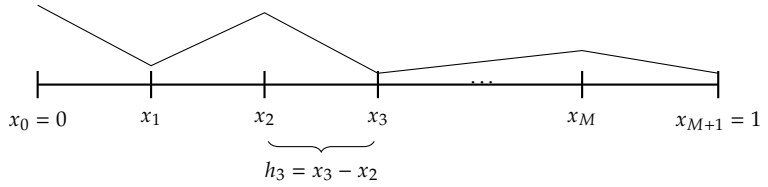
$$g'(0) = \int_0^1 a\,u'v'\,\mathrm{d}x - \int_0^1 fv\,\mathrm{d}x = 0, \tag{1.37}$$

which is exactly the variational problem. Since v was arbitrary, the variational problem is completely equivalent to this energy minimisation. □

Our first finite element method

Idea: solve the variational problem (or the Min problem) approximately by replacing V with a finite-dimensional subspace V_h.

Let $\mathcal{T}_h$ be a mesh on $(0,1)$:



Let

$$h_j = x_j - x_{j-1}, \qquad j = 1, 2, \ldots, M+1. \tag{1.38}$$

The simplest $V_h \subset V$ is the **space of continuous piecewise linear polynomials on $\mathcal{T}_h$**. This is called the

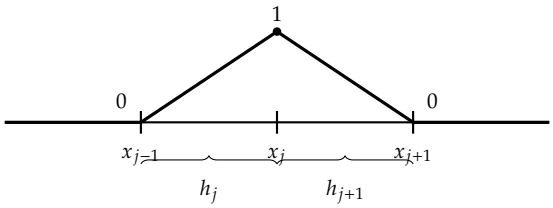
CG(1) method Continuous Galerkin method with polynomials of degree 1. (1.39)

Any function in V_h can be uniquely written as

$$U(x) = \sum_{j=1}^{M} \xi_j \, \varphi_j(x), \tag{1.40}$$

where the ξ_j are coefficients and $\varphi_j(x)$ is the hat function associated with x_j:

$$\varphi_j(x) = \begin{cases} \dfrac{x - x_{j-1}}{h_j}, & x_{j-1} \le x \le x_j, \\ \dfrac{x_{j+1} - x}{h_{j+1}}, & x_j \le x \le x_{j+1}, \\ 0, & \text{everything else, outside those limits.} \end{cases} \tag{1.41}$$



1.4 The discrete system: stiffness matrix and load vector

Lecture 4 — Monday, August 31, 2026 Math 6660 — OneNote export `4th.pdf`

Setup

2-pt BVP. $-(au')' = f$ on $(0,1)$, $u(0) = u(1) = 0$.

$$\Downarrow \text{ variational problem}$$

Find $u \in V$ such that

$$\int_0^1 a\, u'v' \, \mathrm{d}x = \int_0^1 f v \, \mathrm{d}x \qquad \forall\, v \in V, \tag{1.42}$$

where $V = \left\{ v : \int_0^1 v^2 \, \mathrm{d}x < \infty,\ \int_0^1 (v')^2 \, \mathrm{d}x < \infty,\ v(0) = v(1) = 0 \right\}$.

Mesh $\mathcal{T}_h$. Grid points

$$x_0 = 0,\ x_1,\ x_2,\ \ldots,\ x_M,\ x_{M+1} = 1.$$

FEM space.

$$V_h \coloneqq \left\{ v \in V \ :\ v\big|_{[x_i, x_{i+1}]} \in \text{polynomial of degree } 1,\ \ \forall\, i = 0, \ldots, M \right\} \tag{1.43}$$

FEM. Find $U \in V_h \subset V$ such that

$$\int_0^1 a\, U'v' \, \mathrm{d}x = \int_0^1 f v \, \mathrm{d}x \qquad \forall\, v \in V_h. \tag{1.44}$$

A basis for V_h

$\varphi_1(x), \varphi_2(x), \ldots, \varphi_M(x)$, where

$$\varphi_j(x) = \begin{cases} \dfrac{x - x_{j-1}}{h_j}, & x_{j-1} < x < x_j, \\ \dfrac{x_{j+1} - x}{h_{j+1}}, & x_j < x < x_{j+1}, \end{cases} \tag{1.45}$$

$$V_h = \operatorname{span}\{\varphi_1, \varphi_2, \ldots, \varphi_M\}, \qquad \dim V_h = M. \tag{1.46}$$

If $U \in V_h$ then

$$U(x) = \sum_{j=1}^{M} \xi_j \, \varphi_j(x), \qquad \text{where} \qquad \xi_j = U(x_j), \quad j = 1, \ldots, M. \tag{1.47}$$

Galerkin orthogonality

$$\int_0^1 a\, u'v' \,\mathrm{d}x = \int_0^1 f v \,\mathrm{d}x, \qquad u, v \in V, \tag{1.48}$$

$$\int_0^1 a\, U'v' \,\mathrm{d}x = \int_0^1 f v \,\mathrm{d}x, \qquad U, v \in V_h. \tag{1.49}$$

Subtract. Since $V_h \subset V$, (1.48) also holds for $v \in V_h$:

$$\boxed{\int_0^1 a\,(u - U)'\, v' \,\mathrm{d}x = 0 \qquad \text{for all } v \in V_h} \tag{1.50}$$

The linear system

Put (1.47) into (1.44) and take $v = \varphi_i$, $i = 1, \ldots, M$:

$$\int_0^1 a\Big(\sum_{j=1}^{M} \xi_j \varphi_j\Big)' \varphi_i' \,\mathrm{d}x = \int_0^1 f\, \varphi_i \,\mathrm{d}x \quad \Longrightarrow \quad \sum_{j=1}^{M} \xi_j \int_0^1 a\, \varphi_i' \varphi_j' \,\mathrm{d}x = \int_0^1 f\, \varphi_i \,\mathrm{d}x. \tag{1.51}$$

This is an $M \times M$ linear system:

$$\boxed{\boldsymbol{K\xi = b}} \qquad K_{ij} = \int_0^1 a\, \varphi_i' \, \varphi_j' \,\mathrm{d}x, \qquad b_i = \int_0^1 f\, \varphi_i \,\mathrm{d}x, \tag{1.52}$$

with $\boldsymbol{\xi} = (\xi_1, \xi_2, \ldots, \xi_M)^T$.

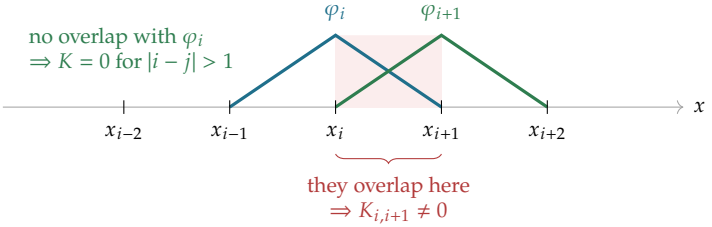


Figure 1.2: Adjacent hats overlap on one element, so only $j = i - 1, i, i + 1$ give a nonzero entry. Sparsity is a consequence of small support, nothing more.

Worked example: $a \equiv 1$, uniform mesh $h_j \equiv h$

Diagonal. $\varphi_i' = +1/h$ on (x_{i-1}, x_i) and $-1/h$ on (x_i, x_{i+1}), so $(\varphi_i')^2 = 1/h^2$ on both:

$$\begin{aligned} K_{ii} &= \int_0^1 1 \cdot \varphi_i' \varphi_i' \, \mathrm{d}x = \int_{x_{i-1}}^{x_i} (\varphi_i')^2 \, \mathrm{d}x + \int_{x_i}^{x_{i+1}} (\varphi_i')^2 \, \mathrm{d}x \\ &= \frac{1}{h^2}(x_i - x_{i-1}) + \frac{1}{h^2}(x_{i+1} - x_i) = \frac{1}{h} + \frac{1}{h} = \frac{2}{h}. \end{aligned} \tag{1.53}$$

Off-diagonal. On (x_i, x_{i+1}): $\varphi_i' = -1/h$ while $\varphi_{i+1}' = +1/h$, so the product is $-1/h^2$ over a length h:

$$K_{i,i+1} = \int_{x_i}^{x_{i+1}} \varphi_i' \, \varphi_{i+1}' \, \mathrm{d}x = -\frac{1}{h^2} \cdot h = -\frac{1}{h}. \tag{1.54}$$

Load vector. By the composite trapezoidal rule on $[x_{i-1}, x_{i+1}]$, using $\varphi_i(x_{i\pm1}) = 0$ and $\varphi_i(x_i) = 1$:

$$b_i = \int_{x_{i-1}}^{x_{i+1}} f \, \varphi_i \, \mathrm{d}x \approx \frac{h}{2}\Big[f(x_{i-1}) \underbrace{\varphi_i(x_{i-1})}_{0} + 2 f(x_i) \underbrace{\varphi_i(x_i)}_{1} + f(x_{i+1}) \underbrace{\varphi_i(x_{i+1})}_{0} \Big] = h \, f(x_i). \tag{1.55}$$

Result.

$$\boldsymbol{K} = \frac{1}{h} \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & \ddots & \ddots & \ddots \\ & & & -1 & 2 \end{bmatrix}, \qquad \boldsymbol{b} = h\big(f(x_1), f(x_2), \ldots, f(x_M)\big)^T. \tag{1.56}$$

Note: this is exactly the central finite difference method

FDM will provide the same matrix as FEM for the 2-pt BVP. Row i of (1.56) reads

$$\frac{1}{h}\big(-\xi_{i-1} + 2\xi_i - \xi_{i+1}\big) = h \, f(x_i) \qquad \Longleftrightarrow \qquad \frac{-\xi_{i-1} + 2\xi_i - \xi_{i+1}}{h^2} = f(x_i), \tag{1.57}$$

which is the $O(h^2)$ central difference approximation of $-u'' = f$.

1.5 Nonhomogeneous Dirichlet and Neumann BCs

Nonhomogeneous Dirichlet BCs

$$-(au')' = f \ \text{ in } (0,1), \qquad u(0) = \alpha, \quad u(1) = \beta. \tag{1.58}$$

Variational problem. Integrating by parts,

$$\int_0^1 au'v' \, \mathrm{d}x - \Big[au'v \Big]_0^1 = \int_0^1 fv \, \mathrm{d}x. \tag{1.59}$$

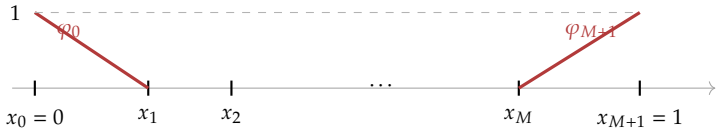
To make the boundary term vanish, define two spaces:

$$V = \Big\{ v : \textstyle\int_0^1 \big[v^2 + (v')^2 \big] \, \mathrm{d}x < \infty, \ \ v(0) = \alpha, \ v(1) = \beta \Big\}, \tag{1.60}$$

$$V_0 = \left\{ v : \int_0^1 \left[v^2 + (v')^2 \right] \mathrm{d}x < \infty, \;\; v(0) = 0 = v(1) \right\}. \tag{1.61}$$

$$\text{Find } u \in V \text{ such that } \int_0^1 a u' v' \, \mathrm{d}x = \int_0^1 f v \, \mathrm{d}x \qquad \forall\, v \in V_0. \tag{1.62}$$

Half-hat functions



Use the same basis $\varphi_j(x)$, $j = 1, \ldots, M$, and introduce two *half-hat* functions:

$$\varphi_0(x) = \frac{x_1 - x}{x_1 - x_0}, \;\; x \in (x_0, x_1); \qquad \varphi_{M+1}(x) = \frac{x - x_M}{x_{M+1} - x_M}, \;\; x \in [x_M, x_{M+1}]. \tag{1.63}$$

Trial function.

$$U(x) = \alpha\, \varphi_0(x) + \beta\, \varphi_{M+1}(x) + \sum_{j=1}^{M} \xi_j\, \varphi_j(x). \tag{1.64}$$

Test functions. $v = \varphi_i(x)$, $i = 1, \ldots, M$.

homogeneous means the boundary is zero: $\alpha = \beta \neq 0$ is *not* homogeneous; $\alpha = \beta = 0$ is homogeneous.

The modified system

Substituting (1.64) into (1.62):

$$\sum_{j=1}^{M} \xi_j \int_0^1 a\, \varphi_i' \varphi_j' \, \mathrm{d}x \;+\; \alpha \int_0^1 a\, \varphi_0' \varphi_i' \, \mathrm{d}x \;+\; \beta \int_0^1 a\, \varphi_{M+1}' \varphi_i' \, \mathrm{d}x \;=\; \int_0^1 f \varphi_i \, \mathrm{d}x. \tag{1.65}$$

Moving the known terms across, and using $\varphi_0' = -1/h_1$, $\varphi_1' = +1/h_1$ on (x_0, x_1):

$$\boxed{\sum_{j=1}^{M} \xi_j \int_0^1 a\, \varphi_i' \varphi_j' \, \mathrm{d}x = \int_0^1 f \varphi_i \, \mathrm{d}x + \delta_{1i} \frac{\alpha}{h_1^2} \int_{x_0}^{x_1} a(x) \, \mathrm{d}x + \delta_{Mi} \frac{\beta}{h_{M+1}^2} \int_{x_M}^{x_{M+1}} a(x) \, \mathrm{d}x} \tag{1.66}$$

Note. Dirichlet BCs are enforced *strongly* and modify the RHS ⇒ no error in the BCs. In FEM language, Dirichlet BCs are called **essential** BCs: they must be imposed in the definition of the function space V.

Neumann BCs

$$-(au')' = f \;\text{ in } (0,1), \qquad \underbrace{u(0) = 0,}_{\text{Dirichlet}} \qquad \underbrace{a(1)u'(1) = g_1}_{\text{Neumann}}. \tag{1.67}$$

Here

$$V = \left\{ v : \int_0^1 \left(v^2 + (v')^2 \right) \mathrm{d}x < \infty, \;\; v(0) = 0 \right\}. \tag{1.68}$$

Only $v(0) = 0$ **is imposed** — nothing is required at $x = 1$.

Integrating by parts:

$$\int_0^1 au'v'\,\mathrm{d}x - \underbrace{a(1)u'(1)}_{=g_1}v(1) + a(0)u'(0)\underbrace{v(0)}_{=0} = \int_0^1 fv\,\mathrm{d}x, \tag{1.69}$$

$$\Longrightarrow \quad \int_0^1 au'v'\,\mathrm{d}x - g_1v(1) = \int_0^1 fv\,\mathrm{d}x \qquad \forall\, v \in V. \tag{1.70}$$

Recovering the BVP (Neumann case)

Q: is the solution of the variational problem also a solution of the 2-pt BVP?
Start from (1.70) and reverse the integration by parts:

$$-\int_0^1 (au')'v\,\mathrm{d}x + a(1)u'(1)v(1) = \int_0^1 fv\,\mathrm{d}x + g_1v(1). \tag{1.71}$$

This must hold for all $v \in V$, *including* all $v \in V$ with $v(1) = 0$. For those,

$$\int_0^1 fv\,\mathrm{d}x = -\int_0^1 (au')'v\,\mathrm{d}x \quad \forall\, v \in V \text{ with } v(1) = 0 \quad \Longrightarrow \quad -(au')' = f \ \text{ in } (0,1). \tag{1.72}$$

Putting that back leaves $a(1)u'(1)v(1) = g_1v(1)$, and since $v(1)$ is arbitrary,

$$\boxed{a(1)u'(1) = g_1} \tag{1.73}$$

CG(1) for the Neumann problem

$$V_h = \{v \in V : v|_{[x_j,x_{j+1}]} = \text{polynomial of degree } 1\} \tag{1.74}$$

$$V_h = \text{span}\{\varphi_1, \varphi_2, \ldots, \varphi_M, \underbrace{\varphi_{M+1}}_{\text{new}}\} \tag{1.75}$$

φ_{M+1} is the half-hat at the right end point. Trial function:

$$U(x) = \sum_{j=1}^{M+1} \xi_j\varphi_j(x), \qquad \int_0^1 aU'v'\,\mathrm{d}x = \int_0^1 fv\,\mathrm{d}x + g_1v(1) \quad \forall\, v \in V_h, \tag{1.76}$$

with $v = \varphi_i$, $i = 1, 2, \ldots, M, \underbrace{M+1}_{\text{new}}$. Note

$$\varphi_i(1) = 0 \ \ \forall\, i = 1, \ldots, M, \qquad \varphi_{M+1}(1) = 1 \ \text{— the only one nonzero at } x = 1. \tag{1.77}$$

At $x = 1$, what is the error? $\Rightarrow$ Not zero.

1.6 Neumann assembly and the first error estimate

Setup, restated

$$-(au')' = f \ \text{ in } (0,1), \qquad \underbrace{u(0) = 0}_{\text{Dirichlet}}, \qquad \underbrace{a(1)u' = g_1}_{\text{Neumann}} \tag{1.78}$$

$$V = \left\{ v : \int \left[v^2 + (v')^2 \right] \mathrm{d}x < \infty, \qquad \underbrace{v(0) = 0}_{\text{strongly enforced}} \right\} \tag{1.79}$$

CG(1) method.

$$U(x) = \sum_{j=1}^{M+1} \xi_j \varphi_j(x) \qquad \text{trial solution; the +1 is because the last node is unknown} \tag{1.80}$$

$$\int_0^1 aU'v' \,\mathrm{d}x = \int_0^1 fv \,\mathrm{d}x + \underbrace{g_1 v(1)}_{\text{Neumann — weakly enforced}}, \qquad v = \varphi_i, \ \ i = 1, 2, \ldots, M+1. \tag{1.81}$$

Example: $a = 1$, uniform mesh $h_i \equiv h$

$$\sum_{j=1}^{M+1} \xi_j \int_0^1 \varphi_i' \varphi_j' \,\mathrm{d}x = \int_0^1 f\varphi_i \,\mathrm{d}x + g_1 \varphi_i(1) \tag{1.82}$$

The new entries, all on the last row and column:

$$A_{(M,M+1)} = \int_{x_M}^{x_{M+1}} \varphi_{M+1}' \, \varphi_M' \,\mathrm{d}x = -\frac{1}{h^2} \int_{x_M}^{x_{M+1}} \mathrm{d}x = -\frac{1}{h}, \tag{1.83}$$

$$A_{(M+1,M)} = -\frac{1}{h} \qquad \Longrightarrow \qquad \text{symmetric matrix}, \tag{1.84}$$

$$A_{(M+1,M+1)} = \int_{x_M}^{x_{M+1}} \left(\varphi_{M+1}' \right)^2 \mathrm{d}x = \frac{1}{h}. \tag{1.85}$$

So $\underline{\underline{A}}\,\underline{\xi} = \underline{b}$ with $\underline{\xi} = (\xi_1, \xi_2, \xi_3, \ldots, \xi_{M+1})$ and

$$A = \frac{1}{h} \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & \boxed{1} \end{bmatrix}, \qquad A \in \mathbb{R}^{(M+1)\times(M+1)}, \ \text{symmetric, positive definite.} \tag{1.86}$$

$$b_i = \int_{x_{i-1}}^{x_{i+1}} f\varphi_i \,\mathrm{d}x \ \ (i = 1, \ldots, M), \qquad b_{M+1} = \int_{x_M}^{x_{M+1}} f\varphi_{M+1} \,\mathrm{d}x + g_1. \tag{1.87}$$

Note: for the 2-pt BVP.

① Dirichlet BCs are **strongly** enforced (essential).

② Neumann or mixed BCs are **weakly** enforced (natural BCs).

Error analysis

Two kinds of error estimates:

① **a priori**: error estimate in terms of derivatives of the exact solution, e.g.

$$\underbrace{\|u\|}_{} \ \dots\ \| \underbrace{u}_{\text{exact}} - \underbrace{U}_{\text{FEA}} \| \leq \underbrace{C}_{\text{constant}} \overbrace{h^{p}}^{\text{order of shape function}} \|u^{(\ell)}\| \qquad \ell^{\text{th}} \text{ derivative of } u.$$

② **a posteriori**: "error estimates in terms of a residual", e.g. $\mathcal{L}(U) - f$. Useful for calculating error *during* the simulation $\Rightarrow$ adaptive error control (adaptive mesh refinement).

Norms

Let $e(x) \coloneqq u(x) - U(x)$ be the error.

$$\text{"Energy" norm:} \qquad \|v\|_E = \sqrt{\int_0^1 a(x)\big[v'(x)\big]^2 \,\mathrm{d}x} \tag{1.88}$$

$$\text{weighted } L_2\text{-norm:} \qquad \|w\|_a = \sqrt{\int_0^1 a(x)\big[w(x)\big]^2 \,\mathrm{d}x} \tag{1.89}$$

$$\textbf{Note: } \|v\|_E = \|v'\|_a. \tag{1.90}$$

Cauchy inequalities.

$$\left| \int_0^1 a v' w' \,\mathrm{d}x \right| \leq \|v'\|_a \|w'\|_a = \|v\|_E \|w\|_E, \tag{1.91}$$

$$\left| \int_0^1 v w \,\mathrm{d}x \right| \leq \|v\|_a \|w\|_{a^{-1}}. \tag{1.92}$$

An a priori error estimate

CG(1) for homogeneous Dirichlet BCs: $-(au')' = f$ in $(0,1)$, $u(0) = u(1) = 0$.

Error calculation.

$$\int_0^1 a U' v' \,\mathrm{d}x = \int_0^1 f v \,\mathrm{d}x \qquad \forall\, v \in V_h \subset V, \tag{1.93}$$

$$\int_0^1 a u' v' \,\mathrm{d}x = \int_0^1 f v \,\mathrm{d}x \qquad \forall\, v \in V. \tag{1.94}$$

Now subtract:

$$\boxed{\int_0^1 a\,(u - U)'\, v' \,\mathrm{d}x = 0 \qquad \forall\, v \in V_h} \qquad \textbf{Galerkin orthogonality} \tag{1.95}$$

Energy norm error estimate. Using (1.90),

$$\|u-U\|_E^2 = \|(u-U)'\|_a^2 = \int_0^1 a\,(u-U)'(u-U)'\,\mathrm{d}x. \tag{1.96}$$

For *any* $v \in V_h$, insert $-v+v$:

$$\begin{aligned}\|u-U\|_E^2 &= \int_0^1 a\,(u-U)'\,(u-v+v-U)'\,\mathrm{d}x \\ &= \int_0^1 a\,(u-U)'(u-v)'\,\mathrm{d}x + \underbrace{\int_0^1 a\,(u-U)'(v-U)'\,\mathrm{d}x}_{=\,0 \text{ by Galerkin orthogonality}}.\end{aligned} \tag{1.97}$$

Note. $u-v \in V$, but $v-U \in V_h$ — which is what lets (1.95) kill the second term.

So, for any $v \in V_h$,

$$\|u-U\|_E^2 = \int_0^1 a\,(u-U)'(u-v)'\,\mathrm{d}x. \tag{1.98}$$

Using Cauchy (1.91),

$$\int_0^1 a\,(u-U)'(u-v)'\,\mathrm{d}x \le \|(u-U)'\|_a\,\|(u-v)'\|_a = \|u-U\|_E\,\|u-v\|_E, \tag{1.99}$$

so $\|u-U\|_E^2 \le \|u-U\|_E\|u-v\|_E$, and cancelling one factor:

$$\boxed{\|u-U\|_E \;\le\; \|u-v\|_E \qquad \forall\, v \in V_h} \tag{1.100}$$

The FE solution U is the best solution, meaning *smallest in the energy norm*.

1.7 A priori error estimates: energy norm and L^2

Setup

$$-(au')' = f \;\text{ in } (0,1), \qquad u(0)=u(1)=0, \tag{1.101}$$

$$V = \Big\{v : \textstyle\int_0^1 \big[v^2+(v')^2\big]\,\mathrm{d}x < \infty,\; v(0)=v(1)=0\Big\}, \qquad V_h \;\text{ for CG(1).} \tag{1.102}$$

(1) Energy norm

$$\|e\|_E^2 = \int_0^1 a\big[(u-U)'\big]^2\,\mathrm{d}x. \tag{1.103}$$

Last time it was shown that the FEM solution satisfies

$$\|u-U\|_E \;\le\; \|u-v\|_E \qquad \forall\, v \in V_h \quad \text{(any other function in } V_h\text{)}. \tag{1.104}$$

Now take the test function

$$v = \Pi_h u \in V_h, \qquad \Pi_h u = \sum_{j=1}^{M} u(x_j)\,\varphi_j(x), \tag{1.105}$$

the projection of $u \in V$ into the subspace $V_h \subset V$: interpolate $u(x)$ onto the mesh using piecewise linear (continuous) functions.

Note.

$$\|u - \Pi_h u\|_E = \|(u - \Pi_h u)'\|_a \leq C \, \|h \, u''\|_a, \tag{1.106}$$

which just comes from approximating u' by a piecewise constant function; C is a constant that depends only on $a(x)$.

Theorem 1.7.1. *The FEM approximation $U \in V_h$ satisfies $\|u - U\|_E \leq \|u - v\|_E$ for all $v \in V_h \subset V$. In particular there exists a constant C, depending only on $a(x)$, such that*

$$\boxed{\|u - U\|_E \leq C \, \|h \, u''\|_a} \tag{1.107}$$

Note. This shows that CG(1) is *first order accurate in the energy norm*, provided $u'' \in L^2(0,1)$.

(2) $L^2(0,1)$ norm

For the same problem (1.101), consider the following **dual problem**:

$$-(a\phi')' = e, \qquad \phi(0) = \phi(1) = 0, \qquad \text{where } e = u - U. \tag{1.108}$$

Note. Because the original 2-pt BVP is self-adjoint, the primal and dual operators are the same.

L^2-norm. Multiply (1.108) by e and integrate:

$$\begin{aligned} \|e\|^2 &= \int_0^1 e^2 \, \mathrm{d}x = -\int_0^1 e \, (a\phi')' \, \mathrm{d}x \\ &= \int_0^1 a \, e' \phi' \, \mathrm{d}x - \Big[a \, e \, \phi' \Big]_0^1, \end{aligned} \tag{1.109}$$

and the boundary term vanishes because $e(0) = e(1) = 0$, so

$$\|e\|^2 = \int_0^1 a \, e' \phi' \, \mathrm{d}x. \tag{1.110}$$

Recall Galerkin orthogonality: $\int_0^1 a \, e' v' \, \mathrm{d}x = 0$ for all $v \in V_h$. Take $v = \Pi_h \phi$:

$$\|e\|^2 = \int_0^1 a \, e' \phi' \, \mathrm{d}x - \int_0^1 a \, e' \, (\Pi_h \phi)' \, \mathrm{d}x = \int_0^1 a \, e' \, (\phi - \Pi_h \phi)' \, \mathrm{d}x. \tag{1.111}$$

Cauchy's inequality.

$$\|e\|^2 \leq \|e'\|_a \, \|(\phi - \Pi_h \phi)'\|_a \tag{1.112}$$

$$\|e\|^2 \leq C_I \, h_{\max} \, \|e'\|_a \underbrace{\|\phi''\|_a}_{\text{interpolation error}} \tag{1.113}$$

But recall, from part (1),

$$\|e'\|_a = \|e\|_E \leq C_I \, h_{\max} \, \|u''\|_a, \tag{1.114}$$

so

$$\|e\|^2 \leq C_I^2 \, h_{\max}^2 \, \|u''\|_a \, \|\phi''\|_a = C_I^2 \, h_{\max}^2 \, \|u''\|_a \, \frac{\|\phi''\|_a}{\|e\|} \, \|e\|, \tag{1.115}$$

and dividing by $\|e\|$,

$$\|e\| \le C_I^2\, h_{\max}^2\, \|u''\|_a\, \frac{\|\phi''\|_a}{\|e\|}. \tag{1.116}$$

Let

$$S_a = \max_{\xi\neq 0} \frac{\|\psi''\|_a}{\|\xi\|}, \qquad \text{where} \qquad -(a\psi')' = \xi, \quad \psi(0)=\psi(1)=0. \tag{1.117}$$

Note. If $a \equiv$ constant, then $-(a\psi')' = -a\psi'' = \xi$, so

$$\frac{\psi''}{\|\xi\|} = \frac{1}{a}. \tag{1.118}$$

1.8 The stability constant, and a posteriori error

Recap

For $-(au')' = f$ on $(0,1)$, $u(0)=u(1)=0$, with CG(1):

(1) Energy norm. $\|u-U\|_E^2 = \int_0^1 a\left[(u-U)'\right]^2 \mathrm{d}x$, and $\|u-U\|_E \le \|u-v\|_E$ for all $v \in V_h$. Taking $v = \Pi_h u$,

$$\|e\|_E \le C\, \|h\, u''\|_a. \tag{1.119}$$

(2) L^2 norm. $\|u-U\|^2 = \int_0^1 (u-U)^2\,\mathrm{d}x$, and the summary bound from last lecture,

$$\|e\| \le C_I^2\, h_{\max}\, \|u''\|_a\, \frac{\|\phi''\|_a}{\|e\|}, \tag{1.120}$$

which still contains the unknown ϕ from the **dual problem**

$$-(a\phi')' = e, \qquad \phi(0)=\phi(1)=0. \tag{1.121}$$

Claim (the stability constant)

Theorem 1.8.1. *For*

$$-(a\psi')' = \xi, \qquad \psi(0)=\psi(1)=0, \tag{1.122}$$

we have

$$\max_{\xi\neq 0} \frac{\|\psi''\|_a}{\|\xi\|} \;\le\; C_S, \tag{1.123}$$

where C_S depends only on a.

Proof. Expanding (1.122), $-a\psi'' = a'\psi' + \xi$, so

$$\|a\psi''\| \le \|\xi\| + \|a'\psi'\| \le \|\xi\| + \|a'\|_\infty \|\psi'\|, \qquad \|a'\|_\infty = \max_{x\in[0,1]} |a'(x)|. \tag{1.124}$$

Take the BVP, multiply by ψ and integrate by parts:

$$-\int_0^1 \psi\,(a\psi')'\,\mathrm{d}x = \int_0^1 \xi\,\psi\,\mathrm{d}x \quad\Longrightarrow\quad \int_0^1 a\,(\psi')^2\,\mathrm{d}x = \int_0^1 \xi\,\psi\,\mathrm{d}x, \tag{1.125}$$

and since $a \ge a_{\min}$,

$$a_{\min}\int_0^1 (\psi')^2\,\mathrm{d}x \le \int_0^1 \xi\,\psi\,\mathrm{d}x \quad\Longrightarrow\quad a_{\min}\|\psi'\|^2 \le \|\xi\|\,\|\psi\|. \tag{1.126}$$

□

Aside: Poincaré inequality

Let $v \in V$. Then

$$v(x) = \int_0^x v'(y)\,\mathrm{d}y, \qquad |v| \le \int_0^1 |v'|\,\mathrm{d}y \le \sqrt{\int_0^1 (v')^2\,\mathrm{d}y}, \tag{1.127}$$

so $\int_0^1 v^2\,\mathrm{d}x \le \int_0^1 (v')^2\,\mathrm{d}y$, i.e.

$$\|v\| \le \|v'\|. \tag{1.128}$$

Back to the proof

From (1.126) and Poincaré (1.128),

$$a_{\min}\|\psi'\|^2 \le \|\xi\|\,\|\psi\| \le \|\xi\|\,\|\psi'\| \qquad \Longrightarrow \qquad \|\psi'\| \le \frac{\|\xi\|}{a_{\min}}. \tag{1.129}$$

Substituting into (1.124),

$$\begin{aligned} \|a\psi''\| &\le \|\xi\| + \|a'\|_\infty \frac{\|\xi\|}{a_{\min}} \\ &\le \left(1 + \frac{\|a'\|_\infty}{a_{\min}}\right)\|\xi\|. \end{aligned} \tag{1.130}$$

But $\sqrt{a_{\min}}\,\|\psi''\|_a \le \|a\psi''\|$, since

$$\|\psi''\|_a = \sqrt{\int_0^1 a\,(\psi'')^2\,\mathrm{d}x} \qquad \text{while} \qquad \|a\psi''\| = \sqrt{\int_0^1 a^2(\psi'')^2\,\mathrm{d}x}. \tag{1.131}$$

Therefore

$$\|\psi''\|_a \le \underbrace{\frac{1 + \dfrac{\|a'\|_\infty}{a_{\min}}}{\sqrt{a_{\min}}}}_{C_S}\|\xi\|, \qquad \text{so} \qquad \frac{\|\psi''\|_a}{\|\xi\|} \le C_S. \tag{1.132}$$

Conclusion of the L^2 estimate

Feeding (1.132) into (1.120),

$$\boxed{\|e\| \le \underbrace{C_I^2 C_S}_{\text{constant}}\|h^2 u''\|_a} \qquad \text{or} \qquad \|e\| \le C_I^2\,C_S\,h_{\max}^2\,\|u''\|_a. \tag{1.133}$$

Assuming $\|u''\|_a$ is bounded, then $\|e\| = O(h^2)$.

A posteriori error

(1) Energy norm.

$$\begin{aligned} \|e\|_E^2 = \|e'\|_a^2 &= \int_0^1 a\,(e')^2\,\mathrm{d}x \\ &= \int_0^1 a\,u'e'\,\mathrm{d}x - \int_0^1 a\,U'e'\,\mathrm{d}x \end{aligned}$$

$$= \int_0^1 f\, e \,\mathrm{d}x - \int_0^1 a\, U' e' \,\mathrm{d}x. \tag{1.134}$$

Subtract from this the discrete equation, which vanishes for any $v \in V_h$:

$$\int_0^1 f\, v \,\mathrm{d}x - \int_0^1 a\, U' v' \,\mathrm{d}x = 0, \qquad \text{with } v = \Pi_h e \in V_h, \tag{1.135}$$

giving

$$\|e\|_E^2 = \int_0^1 f\,\big(e - \Pi_h e\big)\,\mathrm{d}x - \int_0^1 a\, U' \big(e - \Pi_h e\big)' \,\mathrm{d}x. \tag{1.136}$$

1.9 A posteriori error and adaptive refinement

Lecture 9 — Monday, September 14, 2026 Math 6660 — 9th.pdf

Error analysis: where we are

a priori	① energy norm	② L^2 norm
a posteriori	① energy norm	

A posteriori error, energy norm

$$\|e\|_E^2 = \int_0^1 a\, e' e' \,\mathrm{d}x, \tag{1.137}$$

and from the FEM (Lecture 8),

$$\|e\|_E^2 = \int_0^1 f\,\big(e - \Pi_h e\big)\,\mathrm{d}x - \int_0^1 a\, U' \big(e - \Pi_h e\big)' \,\mathrm{d}x. \tag{1.138}$$

But from interpolation,

$$\underbrace{\|h^{-1}\big(e - \Pi_h e\big)\|_a}_{O(h^2)} \;\le\; C\; \underbrace{\|e'\|_a}_{O(h)}. \tag{1.139}$$

Note. Roughly speaking $R(U) = O(1)$.

In practice: the adaptive algorithm

① On some initial coarse mesh, compute the FEM solution.

② Calculate on each element the residual

$$E_j^2 = \big(x_j - x_{j-1}\big)^2 \int_{x_{j-1}}^{x_j} \frac{1}{a(x)}\, R^2(U)\,\mathrm{d}x, \qquad \text{where} \qquad R(U) = f + \big(aU'\big)'. \tag{1.140}$$

We want

$$E^2 = \sum_{j=1}^{M+1} E_j^2 \;\le\; \frac{\epsilon^2}{c^2}, \tag{1.141}$$

which is achieved if each element satisfies

$$E_j^2 \;\le\; \frac{\epsilon^2}{c^2(M+1)}. \tag{1.142}$$

③ Recalculate the solution on the new mesh, then repeat ② + ③.

1.10 CG(2); the beam equation and CG(3)

Lectures 10–11 (combined) Math 6660 — `missing.pdf` (colleague's notes)

These two lectures are missing from my own notebook; the transcription below is from a classmate's notes, so the wording is his, not mine. The material picks up mid-sentence from the adaptive criterion at the end of Lecture 9.

Finishing the adaptive criterion

$$\text{If } E_j^2 \le \frac{\varepsilon^2}{c^2(M+1)} \text{ move to the next element; else refine.} \tag{1.143}$$

CG(2) method

Same problem as before,

$$\begin{cases} -(au')' = f \quad \text{on } (0,1), \\ u(0) = u(1) = 0, \end{cases} \tag{1.144}$$

with the same variational problem

$$\int_0^1 a\,u'v'\,\mathrm{d}x = \int_0^1 f v\,\mathrm{d}x \qquad \forall\, v \in V, \tag{1.145}$$

$$V = \Big\{ v : \int_0^1 \big(v^2 + (v')^2\big)\,\mathrm{d}x < \infty,\ \ v(0) = v(1) = 0 \Big\}. \tag{1.146}$$

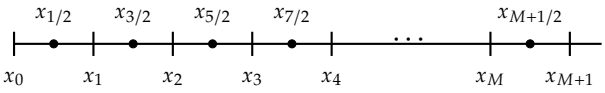
What changes is only the subspace. For CG(2),

$$V_h = \big\{ v \in V : \ v|_{[x_{j-1},x_j]} = a_j + b_j x + c_j x^2 \big\}. \tag{1.147}$$

Key: need to find a suitable basis for V_h.

Mesh $\mathcal{T}_h$: $0 = x_0 < x_1 < \cdots < x_M < x_{M+1} = 1$.

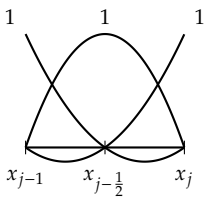
The simplest way to write CG(2) is to use Lagrangian polynomials. To do this, introduce a *submesh*:



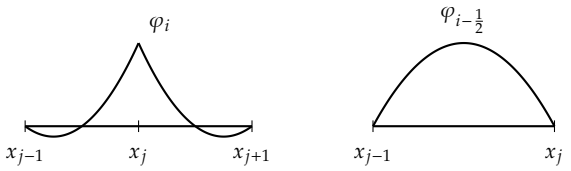
The half-integer nodes need not be the midpoints, but we take them to be.

The basis

For a given element there are 3 Lagrange polynomials:



As *basis functions* over the whole mesh there are two shapes: the one sitting at an integer node, which spans two elements, and the bubble sitting at a half-integer node, which lives inside one element.



Explicitly,

$$\varphi_i(x) = \begin{cases} \dfrac{2\big(x - x_{i+\frac{1}{2}}\big)\big(x - x_{i+1}\big)}{\big(x_{i+1} - x_i\big)^2}, & x_i \le x \le x_{i+1}, \\ \dfrac{2\big(x - x_{i-\frac{1}{2}}\big)\big(x - x_{i-1}\big)}{\big(x_i - x_{i-1}\big)^2}, & x_{i-1} \le x \le x_i, \\ 0, & \text{otherwise,} \end{cases} \tag{1.148}$$

$$\varphi_{i-\frac{1}{2}}(x) = \begin{cases} \dfrac{4\big(x_i - x\big)\big(x - x_{i-1}\big)}{\big(x_i - x_{i-1}\big)^2}, & x_{i-1} \le x \le x_i, \\ 0, & \text{otherwise.} \end{cases} \tag{1.149}$$

Therefore

$$V_h = \operatorname{span}\big\{\varphi_{\frac{1}{2}}, \varphi_1, \varphi_{\frac{3}{2}}, \varphi_2, \dots, \varphi_M, \varphi_{M+\frac{1}{2}}\big\}, \tag{1.150}$$

so if $U \in V_h$ then

$$U(x) = \sum_{j=1}^{2M+1} \xi_j \, \varphi_{j/2}(x). \tag{1.151}$$

Note. For any function $v \in V$,

$$\Pi_h v = \sum_{k=1}^{2M+1} v\big(x_{k/2}\big)\varphi_{k/2}(x), \tag{1.152}$$

and the test functions are $v = \varphi_{i/2}(x)$ for $i = 1, 2, \dots, 2M+1$.

CG(2) variational problem

$$\sum_{j=1}^{2M+1} \xi_j \int_0^1 a(x)\, \varphi'_{i/2}\varphi'_{j/2}\, \mathrm{d}x = \int_0^1 f\, \varphi_{i/2}\, \mathrm{d}x. \tag{1.153}$$

This is a $(2M+1) \times (2M+1)$ symmetric positive definite system, *but not tridiagonal*.

Example: $a \equiv 1$ **and** $h_j = h$ **for all** j

Stiffness matrix.

① $i = 1, 3, \dots, 2M+1$ (the half-integer nodes):

$$A_{ii} = \int_{x_{\frac{i}{2}-\frac{1}{2}}}^{x_{\frac{i}{2}+\frac{1}{2}}} \big(\varphi'_{i/2}\big)^2 \, \mathrm{d}x = \frac{16}{3h}, \tag{1.154}$$

$$A_{(i-1,i)} = A_{(i,i-1)} = \int_{x_{\frac{i-1}{2}}}^{x_{\frac{i+1}{2}}} \varphi'_{i/2}\, \varphi'_{\frac{i-1}{2}}\, \mathrm{d}x = -\frac{8}{3h}. \tag{1.155}$$

② $i = 2, 4, 6, \ldots, 2M$ (the integer nodes):

$$A_{ii} = \int_{x_{\frac{i-1}{2}}}^{x_{\frac{i+1}{2}}} \left[\varphi'_{i/2}\right]^2 \mathrm{d}x = \frac{14}{3h}, \tag{1.156}$$

$$A_{(i,i-1)} = A_{(i-1,i)} = \int_{x_{\frac{i-1}{2}}}^{x_{\frac{i}{2}}} \varphi'_{i/2}\, \varphi'_{\frac{i}{2}-\frac{1}{2}}\, \mathrm{d}x = -\frac{8}{3h}, \tag{1.157}$$

$$A_{(i,i-2)} = A_{(i-2,i)} = \int_{x_{\frac{i-1}{2}}}^{x_{i/2}} \varphi'_{i/2}\, \varphi'_{\frac{i}{2}-1}\, \mathrm{d}x = \frac{1}{3h}. \tag{1.158}$$

RHS vector. Replace f by its interpolant,

$$f \approx \widetilde{\Pi}_h f = \sum_{j=0}^{2M+2} f\left(x_{j/2}\right) \varphi_{j/2}(x). \tag{1.159}$$

① $i = 1, 3, \ldots, 2M+1$:

$$f_i = \sum_{j=1}^{2M+1} f\left(x_{j/2}\right) \int_{x_{\frac{i-1}{2}}}^{x_{\frac{i+1}{2}}} \varphi_{j/2}\, \varphi_{i/2}\, \mathrm{d}x = \frac{h}{15}\left[f\left(x_{\frac{i-1}{2}}\right) + 8f\left(x_{i/2}\right) + f\left(x_{\frac{i+1}{2}}\right)\right]. \tag{1.160}$$

② $i = 2, 4, \ldots, 2M$:

$$f_i = \frac{h}{30}\left[-f\left(x_{\frac{i-2}{2}}\right) + 2f\left(x_{\frac{i-1}{2}}\right) + 8f\left(x_{i/2}\right) + 2f\left(x_{\frac{i+1}{2}}\right) - f\left(x_{\frac{i+2}{2}}\right)\right]. \tag{1.161}$$

Therefore

$$\boldsymbol{A} = \frac{1}{3h}\begin{bmatrix} 16 & -8 & & & & & & \\ -8 & 14 & -8 & 1 & & & & \\ & -8 & 16 & -8 & & & & \\ & 1 & -8 & 14 & -8 & 1 & & \\ & & & \ddots & \ddots & \ddots & & \\ & & & 1 & -8 & 14 & -8 & \\ & & & & & -8 & 16 & \end{bmatrix}, \qquad \boldsymbol{Ax} = \boldsymbol{b}. \tag{1.162}$$

Note (energy norm error). Galerkin orthogonality gives $\|u - U\|_E \le \|u - v\|_E$ for all $v \in V_h$; taking $v = \Pi_h u$,

$$\boxed{\|u - U\|_E \le c\, \|h^2 u'''\|} \tag{1.163}$$

Note (L^2 error).

$$\boxed{\|u - U\| \le c\, \|h^3 u'''\|} \tag{1.164}$$

Beam equation

$$\begin{cases} u^{(iv)} = f & \text{on } 0 < x < 1, \\ u(0) = u'(0) = u(1) = u'(1) = 0. \end{cases} \tag{1.165}$$

Variational problem

Multiply by v and integrate by parts twice:

$$\int_0^1 v\, u^{(iv)}\,\mathrm{d}x = \int_0^1 f v\,\mathrm{d}x, \tag{1.166}$$

$$-\int_0^1 v' u'''\,\mathrm{d}x + \underbrace{\left[v\,u'''\right]_0^1}_{=0} = \int_0^1 f v\,\mathrm{d}x \qquad (\text{let } v(0)=v(1)=0), \tag{1.167}$$

$$\int_0^1 v'' u''\,\mathrm{d}x - \underbrace{\left[v' u''\right]_0^1}_{=0} = \int_0^1 f v\,\mathrm{d}x \qquad (\text{let } v'(0)=v'(1)=0), \tag{1.168}$$

so

$$\int_0^1 u'' v''\,\mathrm{d}x = \int_0^1 f v\,\mathrm{d}x. \tag{1.169}$$

The space is

$$V = \left\{ v : \int_0^1 \left[v^2 + (v')^2 + (v'')^2\right]\mathrm{d}x < \infty,\ \ v(0)=v(1)=v'(0)=v'(1)=0 \right\}, \tag{1.170}$$

and functions in V are $C^1[0,1]$.

(Variational problem) Find $u \in V$ such that $\displaystyle\int_0^1 u'' v''\,\mathrm{d}x = \int_0^1 f v\,\mathrm{d}x$ for all $v \in V$.

Energy minimization

Define

$$F(w) = \frac{1}{2}\int_0^1 (w'')^2\,\mathrm{d}x - \int_0^1 f w\,\mathrm{d}x. \tag{1.171}$$

(Minimization problem) Find $u \in V$ such that $F(u) \le F(w)$ for all $w \in V$.

Claim. (Var. Prob.) $\Longleftrightarrow$ (Min. Prob.)

($\Longrightarrow$) Assume $u \in V$ satisfies the variational problem. Let $w = u + v \in V$. Then

$$F(w) = F(u+v) = \underbrace{\frac{1}{2}\int_0^1 (u'')^2\,\mathrm{d}x - \int_0^1 f u\,\mathrm{d}x}_{=F(u)} + \frac{1}{2}\int_0^1 (v'')^2\,\mathrm{d}x + \underbrace{\int_0^1 u'' v''\,\mathrm{d}x - \int_0^1 f v\,\mathrm{d}x}_{=0 \text{ by } (1.169)}, \tag{1.172}$$

so

$$F(w) = F(u) + \frac{1}{2}\int_0^1 (v'')^2\,\mathrm{d}x \ \ge\ F(u). \tag{1.173}$$

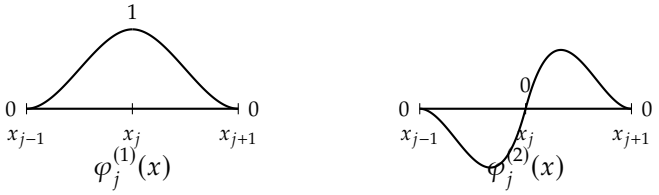
Note. $F(w) = F(u) \ \Rightarrow\ v'' \equiv 0 \ \Rightarrow\ v \equiv 0$ (due to the BCs) $\Rightarrow\ u \equiv w$.

CG(3) method

We need $V_h \subseteq V$. On each element $[x_{j-1}, x_j]$ we need a polynomial that matches *value and derivative* to its neighbours at x_{j-1} and x_j — that is 4 conditions, so we need a cubic (4 coefficients):

$$V_h = \left\{ v \in V : \; v|_{[x_{j-1},x_j]} = a_j + b_j x + c_j x^2 + d_j x^3 \right\}. \tag{1.174}$$

Key: find a basis for V_h. Two basis functions are attached to each node — one controlling the value, one controlling the slope:



They are fixed by the nodal table

	x_{j-1}	x_j	x_{j+1}
$\varphi_j^{(1)}$	0	1	0
$(\varphi_j^{(1)})'$	0	0	0
$\varphi_j^{(2)}$	0	0	0
$(\varphi_j^{(2)})'$	0	1	0

which gives

$$\varphi_j^{(1)} = \begin{cases} 3\left(\dfrac{x - x_{j-1}}{x_j - x_{j-1}}\right)^2 - 2\left(\dfrac{x - x_{j-1}}{x_j - x_{j-1}}\right)^3, & x_{j-1} \le x \le x_j, \\ 3\left(\dfrac{x - x_{j+1}}{x_j - x_{j+1}}\right)^2 - 2\left(\dfrac{x - x_{j+1}}{x_j - x_{j+1}}\right)^3, & x_j \le x \le x_{j+1}, \\ 0, & \text{otherwise}, \end{cases} \tag{1.175}$$

$$\varphi_j^{(2)} = \begin{cases} -\dfrac{1}{h_j}(x - x_{j-1})^2 + \dfrac{1}{h_j^2}(x - x_{j-1})^3, & x_{j-1} \le x \le x_j, \\ \dfrac{1}{h_{j+1}}(x - x_{j+1})^2 + \dfrac{1}{h_{j+1}^2}(x - x_{j+1})^3, & x_j \le x \le x_{j+1}, \\ 0, & \text{otherwise}. \end{cases} \tag{1.176}$$

CG(3) FEM: Find $U \in V_h$ such that $\displaystyle\int_0^1 U'' v'' \, \mathrm{d}x = \int_0^1 f v \, \mathrm{d}x$ for all $v \in V_h$.

1.11 CG(3) assembly: the Hermite stiffness matrix

Lecture 12 — Wednesday, September 23, 2026 Math 6660 — `12th.pdf`

Lecture 12 re-states the beam problem from Lecture 11 and then carries the CG(3) discretisation all the way to the assembled linear system. Pages 7–9 of `missing.pdf` *cover the same material; the two sources agree throughout, and the transcription below follows my own notes.*

Recap: the beam problem and its CG(3) space

$$\begin{cases} u^{(iv)} = f & \text{in } (0,1), \\ u(0) = u(1) = u'(0) = u'(1) = 0, \end{cases} \tag{1.177}$$

$$V = \left\{ v : \int_0^1 \left[v^2 + (v')^2 + (v'')^2 \right] \mathrm{d}x < \infty, \;\; v(0) = v(1) = v'(0) = v'(1) = 0 \right\}. \tag{1.178}$$

Functions in V are C^1. Find $u \in V$ such that

$$\int_0^1 u''v'' \,\mathrm{d}x = \int_0^1 f v \,\mathrm{d}x \qquad \forall\, v \in V. \tag{1.179}$$

FEM: $V_h \subset V$, spanned by the two Hermite shapes per node — $\varphi_j^{(1)}$, which carries the nodal value, and $\varphi_j^{(2)}$, which carries the nodal slope (Lecture 11, (1.175)–(1.176)):

$$V_h = \operatorname{span}\{\varphi_1^{(1)}, \ldots, \varphi_M^{(1)}, \; \varphi_1^{(2)}, \ldots, \varphi_M^{(2)}\}, \qquad \dim V_h = 2M. \tag{1.180}$$

Trial functions:

$$U = \sum_{j=1}^{M} \xi_j \, \varphi_j^{(1)}(x) + \sum_{j=1}^{M} \eta_j \, \varphi_j^{(2)}(x). \tag{1.181}$$

Test functions: $v = \varphi_1^{(1)}, \ldots, \varphi_M^{(1)}, \varphi_1^{(2)}, \ldots, \varphi_M^{(2)}$.

The interpolant and its errors

If $u \in V$, then

$$\Pi_h u = \sum_{j=1}^{M} u(x_j) \, \varphi_j^{(1)}(x) + \sum_{j=1}^{M} u'(x_j) \, \varphi_j^{(2)}(x), \qquad \text{where } \Pi_h u \in V_h. \tag{1.182}$$

Interpolation errors:

$$\|u - \Pi_h u\|_{L^2(0,1)} \le c_1 \|h^4 u^{(4)}\|_{L^2(0,1)}, \tag{1.183}$$

$$\|(u - \Pi_h u)'\|_{L^2(0,1)} \le c_2 \|h^3 u^{(4)}\|_{L^2(0,1)}, \tag{1.184}$$

$$\|(u - \Pi_h u)''\|_{L^2(0,1)} \le c_3 \|h^2 u^{(4)}\|_{L^2(0,1)}. \tag{1.185}$$

Example: uniform mesh, $h_j = h$ for all j

With

$$\int_0^1 U''v'' \,\mathrm{d}x = \int_0^1 f v \,\mathrm{d}x, \qquad v = \varphi_1^{(1)}, \ldots, \varphi_M^{(1)}, \varphi_1^{(2)}, \ldots, \varphi_M^{(2)}, \tag{1.186}$$

the stiffness matrix comes out of four families of integrals. Testing against $\varphi_k^{(1)}$, $k = 1, 2, \ldots, M$:

$$\int_0^1 U''v'' \,\mathrm{d}x = \sum_{j=1}^{M} \xi_j \int_0^1 (\varphi_j^{(1)})'' (\varphi_k^{(1)})'' \,\mathrm{d}x + \sum_{j=1}^{M} \eta_j \int_0^1 (\varphi_j^{(2)})'' (\varphi_k^{(1)})'' \,\mathrm{d}x, \tag{1.187}$$

and then again, testing against $\varphi_k^{(2)}$, $k = 1, 2, \ldots, M$:

$$\sum_{j=1}^{M} \xi_j \int_0^1 (\varphi_j^{(1)})'' (\varphi_k^{(2)})'' \,\mathrm{d}x + \sum_{j=1}^{M} \eta_j \int_0^1 (\varphi_j^{(2)})'' (\varphi_k^{(2)})'' \,\mathrm{d}x. \tag{1.188}$$

That is a block matrix

$$\left[\begin{array}{c|c} \boldsymbol{A} & \boldsymbol{C} \\ \hline -\boldsymbol{C} & \boldsymbol{B} \end{array}\right] \in \mathbb{R}^{2M \times 2M}, \qquad \text{here } \boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{R}^{M \times M}. \tag{1.189}$$

The three blocks

Now,

$$A_{jk} = \int_0^1 \big(\varphi_j^{(1)}\big)''\big(\varphi_k^{(1)}\big)'' \,\mathrm{d}x, \qquad \boldsymbol{A} = \frac{12}{h^3}\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & 2 \end{bmatrix}, \tag{1.190}$$

which is $M \times M$, tridiagonal, symmetric.

$$B_{jk} = \int_0^1 \big(\varphi_j^{(2)}\big)''\big(\varphi_k^{(2)}\big)'' \,\mathrm{d}x, \qquad \boldsymbol{B} = \frac{2}{h}\begin{bmatrix} 4 & 1 & & \\ 1 & 4 & \ddots & \\ & \ddots & \ddots & 1 \\ & & 1 & 4 \end{bmatrix}, \tag{1.191}$$

which is $M \times M$, tridiagonal, symmetric.

$$C_{jk} = \int_0^1 \big(\varphi_j^{(1)}\big)''\big(\varphi_k^{(2)}\big)'' \,\mathrm{d}x, \qquad \boldsymbol{C} = \frac{6}{h^2}\begin{bmatrix} 0 & 1 & & \\ -1 & 0 & \ddots & \\ & \ddots & \ddots & 1 \\ & & -1 & 0 \end{bmatrix}, \tag{1.192}$$

which is $M \times M$, tridiagonal — but *not* symmetric. It is anti-symmetric:

$$\boldsymbol{C}^T = -\boldsymbol{C}. \tag{1.193}$$

Therefore

$$\boldsymbol{S} = \left[\begin{array}{c|c} \boldsymbol{A} & \boldsymbol{C} \\ \hline \boldsymbol{C}^T & \boldsymbol{B} \end{array}\right], \qquad \boldsymbol{S}^T = \boldsymbol{S}, \tag{1.194}$$

so $\boldsymbol{S}$ is $2M \times 2M$ and symmetric.

RHS

Interpolate the load with the same Hermite basis,

$$f \;\longrightarrow\; \widetilde{\Pi}_h f = \sum_{j=0}^{M+1} f(x_j)\,\varphi_j^{(1)}(x) + \sum_{j=0}^{M+1} f'(x_j)\,\varphi_j^{(2)}(x), \tag{1.195}$$

$$\int_0^1 f v \,\mathrm{d}x \;\longrightarrow\; \int_0^1 \big(\widetilde{\Pi}_h f\big) v \,\mathrm{d}x. \tag{1.196}$$

Now, for $j = 1, 2, \ldots, M$,

$$f_j^{(1)} = \frac{h}{70}\Big[9f(x_{j-1}) + 52f(x_j) + 9f(x_{j+1})\Big] + \frac{13h^2}{420}\Big[f'(x_{j-1}) - f'(x_{j+1})\Big], \tag{1.197}$$

$$f_j^{(2)} = \frac{13h^2}{420}\Big[f(x_{j+1}) - f(x_{j-1})\Big] - \frac{h^3}{420}\Big[3f'(x_{j-1}) - 8f'(x_j) + 3f'(x_{j+1})\Big]. \tag{1.198}$$

The final system

$$\left[\begin{array}{c|c} \boldsymbol{A} & \boldsymbol{C} \\ \hline -\boldsymbol{C} & \boldsymbol{B} \end{array}\right]\left[\begin{array}{c} \boldsymbol{\xi} \\ \hline \boldsymbol{\eta} \end{array}\right] = \left[\begin{array}{c} \boldsymbol{f}^{(1)} \\ \hline \boldsymbol{f}^{(2)} \end{array}\right], \tag{1.199}$$

which is $2M \times 2M$, symmetric, positive definite. Here $\boldsymbol{\eta}$ is the vector of *derivative values at the nodes*.

1.12 The abstract framework: Lax–Milgram, Céa, and the move to 2D

Lectures 13–18 — late September to mid-October, 2026 Math 6660 — `prashant2.pdf` (colleague's notes)

*These lectures are missing from my own notebook; the transcription below follows a classmate's notes, so the wording and the order of presentation are his, not mine. Only two dates are legible in the scan — **2/Oct/26** at the abstract variational formulation, and **12/Oct/26** at the biharmonic example — so the lecture numbering above is my reconstruction, not something written in the source. Pages 1–18 of that scan repeat material already transcribed in Lectures 7–12; the transcription starts where the new material does.*

Part I — the abstract formulation

So far: two examples

Both problems solved so far have the same shape.

(1) The two-point BVP.

$$-(au')' = f \ \text{ in } (0,1), \qquad u(0) = u(1) = 0, \tag{1.200}$$

$$V = \left\{ v : \int_0^1 \left[v^2 + (v')^2 \right] \mathrm{d}x < \infty, \ \ v(0) = v(1) = 0 \right\}, \tag{1.201}$$

$$B(u,v) = \int_0^1 a\, u'v' \, \mathrm{d}x, \qquad L(v) = \int_0^1 f v \, \mathrm{d}x. \tag{1.202}$$

(2) The beam equation.

$$V = \left\{ v : \int_0^1 \left[v^2 + (v')^2 + (v'')^2 \right] \mathrm{d}x < \infty, \ \ v(0) = v(1) = v'(0) = v'(1) = 0 \right\}, \tag{1.203}$$

$$B(u,v) = \int_0^1 u''v'' \, \mathrm{d}x, \qquad L(v) = \int_0^1 f v \, \mathrm{d}x. \tag{1.204}$$

In both cases:

$$\textbf{(Var. Prob.)} \quad \text{Find } u \in V \text{ s.t. } B(u,v) = L(v) \quad \forall\, v \in V, \tag{1.205}$$

$$F(w) = \tfrac{1}{2} B(w,w) - L(w), \qquad \textbf{(Energy Min.)} \quad \text{Find } u \in V \text{ s.t. } F(u) \le F(w) \ \ \forall\, w \in V. \tag{1.206}$$

Note. We seek a **conforming** FEM: $V_h \subset V$. (A non-conforming FEM has $V_h \not\subset V$.)

Multi-index notation

Consider a domain $\Omega \subset \mathbb{R}^d$. Derivatives are defined via a **multi-index**: let $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_d)$ be a d-tuple, where the α_i are non-negative integers. The *length* of α is

$$|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_d, \tag{1.207}$$

and the corresponding derivative of v is

$$D^\alpha v = \frac{\partial^{|\alpha|} v}{\partial x_1^{\alpha_1} \, \partial x_2^{\alpha_2} \cdots \partial x_d^{\alpha_d}}. \tag{1.208}$$

Ex. If $d = 2$ and $|\alpha| = 2$:

$$D^{(2,0)} v = \frac{\partial^2 v}{\partial x_1^2}, \qquad D^{(0,2)} v = \frac{\partial^2 v}{\partial x_2^2}, \qquad D^{(1,1)} v = \frac{\partial^2 v}{\partial x_1 \partial x_2}.$$

Sobolev spaces

$$L^1_{\text{loc}}(\Omega) \coloneqq \big\{v : \ v \in L^1(K) \text{ for any compact subset } K \text{ inside } \Omega\big\}. \tag{1.209}$$

Defn (Sobolev space).

$$W^{r,q}(\Omega) = \Big\{v \in L^1_{\text{loc}}(\Omega) : \ \|v\|_{W^{r,q}(\Omega)} < \infty\Big\}, \tag{1.210}$$

where

$$\|v\|_{W^{r,q}(\Omega)} \coloneqq \Big[\sum_{|\alpha|\le r} \|D^\alpha v\|^q_{L^q(\Omega)}\Big]^{1/q}, \qquad 1 \le q < \infty, \quad r \ge 0, \tag{1.211}$$

and

$$\|v\|^q_{L^q(\Omega)} = \int_\Omega |v|^q \,\mathrm{d}x. \tag{1.212}$$

Defn (Sobolev space with homogeneous BCs).

$$W^{r,q}_0(\Omega) = \Big\{v \in W^{r,q}(\Omega) : \ D^\alpha v\big|_{\partial\Omega} = 0 \text{ for } |\alpha| < r\Big\}. \tag{1.213}$$

Hilbert spaces

Defn (Hilbert space). When $q = 2$ we write $W^{r,2}(\Omega) = H^r(\Omega)$. This is called a **Hilbert space**, with

$$\|v\|_{H^r(\Omega)} = \sqrt{(v,v)_{H^r(\Omega)}}, \qquad (v,w)_{H^r(\Omega)} \coloneqq \sum_{|\alpha|\le r} \int_\Omega \big(D^\alpha v\big)\big(D^\alpha w\big)\,\mathrm{d}x. \tag{1.214}$$

Note. $r = 0$ also makes sense: $H^0(\Omega) = L^2(\Omega)$.

Defn (semi-norm).

$$|v|_{H^r(\Omega)} = \sqrt{\sum_{|\alpha|=r} \int_\Omega \big(D^\alpha v\big)^2 \,\mathrm{d}x}. \tag{1.215}$$

Note. For the remainder of the discussion, let V be a Hilbert space with inner product $(\cdot,\cdot)_V$ and corresponding norm $\|\cdot\|_V$.

Linear functionals and the dual space

Defn. A mapping $L : V \to \mathbb{R}$ is a **linear functional** if

$$L(\alpha u + \beta v) = \alpha L(u) + \beta L(v) \qquad \forall\, \alpha,\beta \in \mathbb{R},\ \forall\, u, v \in V. \tag{1.216}$$

L is **bounded** in the norm $\|\cdot\|_V$ if there exists a constant $\widetilde{L} > 0$ such that

$$|L(v)| \le \widetilde{L}\,\|v\|_V \qquad \forall\, v \in V. \tag{1.217}$$

Note. A bounded linear functional is continuous:

$$L(v) - L(w) = L(v-w) \implies |L(v) - L(w)| = |L(v-w)| \le \widetilde{L}\|v-w\|_V.$$

The converse is also true: a continuous linear functional is bounded.

Defn. The set of bounded linear functionals on V is termed the **dual space** of V, denoted V', equipped with the norm

$$\|L\|_{V'} = \sup_{\substack{v\in V\\ v\neq 0}} \frac{L(v)}{\|v\|_V} \ \le\ \widetilde{L}. \tag{1.218}$$

Ex. (dual space of $L^q(\Omega)$). Consider the linear functional $L(v) = \int_\Omega f(x)v(x)\,\mathrm{d}x$ for $f \in L^{q'}(\Omega)$ where $\frac{1}{q} + \frac{1}{q'} = 1$. **Hölder's inequality**,

$$\|uv\|_{L^1(\Omega)} \le \|u\|_{L^{q'}(\Omega)}\|v\|_{L^q(\Omega)} \qquad \forall\, u \in L^{q'}(\Omega),\ v \in L^q(\Omega), \tag{1.219}$$

gives $|L(v)| \le \|f\|_{L^{q'}}\|v\|_{L^q}$. So every $f \in L^{q'}(\Omega)$ can be viewed as a bounded linear functional on $L^q(\Omega)$; every bounded linear functional on $L^q(\Omega)$ is of this form; hence $L^{q'}(\Omega)$ is the dual space of $L^q(\Omega)$.

Thm (Riesz representation). Let H be a Hilbert space with scalar product $(\cdot,\cdot)_H$. Then for any continuous linear functional $L : H \to \mathbb{R}$ there exists a *unique* $u \in H$ such that

$$L(v) = (u, v)_H \qquad \forall\, v \in H. \tag{1.220}$$

Abstract variational problems

Defn (bilinear form). Let V be a Hilbert space with scalar product $(\cdot,\cdot)_V$. Then $B(\cdot,\cdot) : V \times V \to \mathbb{R}$ is a **bilinear form** if

$$B(u, \alpha v + \beta w) = \alpha B(u, v) + \beta B(u, w), \qquad B(\alpha u + \beta v, w) = \alpha B(u, w) + \beta B(v, w) \tag{1.221}$$

for all $\alpha, \beta \in \mathbb{R}$ and all $u, v, w \in V$.

The three properties that everything below depends on:

(S) Symmetric. $B(u, v) = B(v, u) \quad \forall\, u, v \in V$.

(CB) Continuous / bounded. There exists $\beta^* > 0$ such that

$$B(u, v) \le |B(u, v)| \le \beta^*\|u\|_V\|v\|_V \qquad \forall\, u, v \in V.$$

(VE) V-elliptic (a.k.a. coercive). There exists $\beta_* > 0$ such that

$$B(v, v) \ge \beta_*\|v\|_V^2 \qquad \forall\, v \in V.$$

Defn (abstract variational problem). Let $B(\cdot,\cdot) : V \times V \to \mathbb{R}$ be a bilinear form satisfying (S), (CB) and (VE). Let $L(\cdot) : V \to \mathbb{R}$ be a linear functional bounded in $\|\cdot\|_V$, i.e. $|L(v)| \le \widetilde{L}\|v\|_V\ \forall v \in V$. Then

$$\textbf{(1) Var. Prob.} \quad \text{Find } u \in V \text{ s.t. } B(u, v) = L(v)\ \ \forall\, v \in V, \tag{1.222}$$

$$\textbf{(2) Energy Min.} \quad \text{Find } u \in V \text{ s.t. } F(u) \le F(w)\ \ \forall\, w \in V, \quad F(v) = \tfrac{1}{2}B(v, v) - L(v). \tag{1.223}$$

Claim: (1) and (2) are equivalent

($\Longrightarrow$) Assume $u \in V$ satisfies the variational problem. Let $w = u + v \in V$ for some $v \in V$. Then

$$\begin{aligned} F(w) = F(u + v) &= \tfrac{1}{2}B(u + v, u + v) - L(u + v), \\ B(u + v, u + v) &= B(u, u + v) + B(v, u + v) = B(u, u) + 2B(u, v) + B(v, v) \quad \text{(used (S))}, \end{aligned} \tag{1.224}$$

so

$$F(u + v) = \underbrace{\left[\tfrac{1}{2}B(u, u) - L(u)\right]}_{F(u)} + \underbrace{\left[B(u, v) - L(v)\right]}_{=0 \text{ by (Var. Prob.)}} + \tfrac{1}{2}B(v, v), \tag{1.225}$$

i.e. $F(w) = F(u) + \frac{1}{2}B(v, v)$. But $B(v, v) \ge \beta_*\|v\|_V^2 > 0$ for $v \ne 0$ by (VE), hence

$$F(u) \le F(w) \qquad \forall\, w \in V. \tag{1.226}$$

($\Longleftarrow$) Assume $u \in V$ satisfies $F(u) \le F(w)\ \forall w \in V$. Let

$$g(\varepsilon) = F(u + \varepsilon v) \qquad \text{for fixed but arbitrary } v \in V. \tag{1.227}$$

Expanding,

$$g(\varepsilon) = F(u) + \varepsilon B(u,v) - \varepsilon L(v) + \tfrac{\varepsilon^2}{2} B(v,v), \tag{1.228}$$

$$g'(\varepsilon) = B(u,v) - L(v) + \varepsilon\, B(v,v), \tag{1.229}$$

and $g'(0) = 0$ gives $B(u,v) = L(v)$. □

Recall (this is Lecture 3 and Lecture 11 again). The bar version of this proof is Lecture 3; the beam version is Lecture 11. The algebra is identical in all three — complete the square one way, take $\frac{\mathrm{d}}{\mathrm{d}\varepsilon}$ at $\varepsilon = 0$ the other. What is new is only that it is now done *once*, for every problem of this shape at the same time.

The Lax–Milgram theorem [2/Oct/26]

Thm (Lax–Milgram). Let V be a Hilbert space with norm $\|\cdot\|_V$. Let $B(\cdot,\cdot) : V \times V \to \mathbb{R}$ be a bilinear form satisfying (S), (CB) and (VE). Let $L(\cdot) : V \to \mathbb{R}$ be a bounded linear functional. Then there exists a **unique** solution to the variational problem

$$\text{find } u \in V \text{ s.t. } B(u,v) = L(v) \ \ \forall\, v \in V,$$

with the property that

$$\|u\|_V \le \frac{\widetilde{L}}{\beta_*},$$

where $\widetilde{L}, \beta_*$ are real positive constants.

Proof. Since $B(\cdot,\cdot)$ is symmetric and V-elliptic, it induces a scalar product on V:

$$[u,v] \coloneqq B(u,v) \qquad \forall\, u, v \in V. \tag{1.230}$$

By (VE) and (CB),

$$[v,v] = B(v,v) \ge \beta_* \|v\|_V^2, \qquad [v,v] = B(v,v) \le \beta^* \|v\|_V^2, \tag{1.231}$$

so

$$\boxed{\beta_* \|v\|_V^2 \le [v,v] \le \beta^* \|v\|_V^2} \tag{1.232}$$

and the norms $\sqrt{[v,v]}$ and $\|v\|_V$ are **equivalent**. Therefore $L(\cdot)$ is bounded in both. By the Riesz representation theorem (1.220) there exists a unique $u \in V$ such that $L(v) = [u,v]\ \forall v \in V$, i.e. a unique $u \in V$ with $B(u,v) = L(v)\ \forall v \in V$.

For the solution bound,

$$\beta_* \|u\|_V^2 \le B(u,u) = L(u) \le \widetilde{L}\|u\|_V \implies \|u\|_V \le \frac{\widetilde{L}}{\beta_*}. \quad \square \tag{1.233}$$

Part II — conforming FEM in the abstract setting

The discrete problem

Replace V by a finite-dimensional subspace $V_h \subset V$ (*conforming*):

$$\text{Find } u_h \in V_h \text{ s.t. } B(u_h, v) = L(v) \quad \forall\, v \in V_h. \tag{1.234}$$

Lax–Milgram still applies to this, so there exists a unique solution with $\|u_h\|_V \le \widetilde{L}/\beta_*$.

Let $\{\varphi_i\}_{i=1}^M$ be a basis for V_h, take $v = \varphi_j$, $j = 1, \dots, M$, and write $u_h = \sum_{i=1}^M \xi_i \varphi_i$. Then

$$\sum_{i=1}^{M} B(\varphi_i, \varphi_j)\, \xi_i = L(\varphi_j), \qquad j = 1, 2, \dots, M, \tag{1.235}$$

i.e. the linear system

$$\boxed{\boldsymbol{A}\,\boldsymbol{\xi} = \boldsymbol{b}} \qquad A_{ij} = B(\varphi_i, \varphi_j), \quad \boldsymbol{\xi} = (\xi_1, \xi_2, \dots, \xi_M), \quad b_j = L(\varphi_j). \tag{1.236}$$

Claim: under (S) and (VE), $\boldsymbol{A}$ is symmetric positive definite

(1) Symmetry follows directly from (S):

$$A_{ij} = B(\varphi_i, \varphi_j) = B(\varphi_j, \varphi_i) = A_{ji}. \tag{1.237}$$

(2) Positive definiteness. Let $w = \sum_{i=1}^M w_i \varphi_i \in V_h$. By (VE), $B(w, w) \ge \beta_* \|w\|_V^2$, so if $w \not\equiv 0$ then $B(w, w) > 0$. Expanding,

$$B\Big(\sum_{i=1}^{M} w_i \varphi_i,\, w\Big) = \sum_{i=1}^{M} w_i B(\varphi_i, w) = \sum_{i=1}^{M} w_i B\Big(\varphi_i, \sum_{j=1}^{M} w_j \varphi_j\Big) = \sum_{i=1}^{M}\sum_{j=1}^{M} w_i \underbrace{B(\varphi_i, \varphi_j)}_{A_{ij}} w_j = \boldsymbol{w}^T \boldsymbol{A}\, \boldsymbol{w} > 0 \tag{1.238}$$

by (VE). Hence $\boldsymbol{A}$ is positive definite. □

Note. This means a conforming FEM for this abstract variational problem will generate an SPD linear system — *independent of* the number of dimensions, the polynomial order, the shape of the solution domain Ω, the type of mesh, etc.

Céa's lemma

Thm (Céa's lemma). Let V be a Hilbert space with norm $\|\cdot\|_V$, let B be a bilinear form satisfying (S), (CB) and (VE), and let L be a linear bounded functional. Let $V_h \subset V$ be a finite-dimensional subspace, and let u, u_h solve (1.222) and (1.234) respectively. Then

$$\|u - u_h\|_V \;\le\; \frac{\beta^*}{\beta_*}\, \|u - v\|_V \qquad \forall\, v \in V_h.$$

Proof. Subtract (Var. Prob.) and (FEM):

$$B(u, v) - B(u_h, v) = 0 \;\; \forall\, v \in V_h \qquad \Longrightarrow \qquad \boxed{B(u - u_h,\, v) = 0 \quad \forall\, v \in V_h} \tag{1.239}$$

which is **Galerkin orthogonality**. Now take

$$B(u - u_h,\, u - u_h) = B(u - u_h,\, u - v + v - u_h) = B(u - u_h,\, u - v) + \underbrace{B(u - u_h,\, v - u_h)}_{=0 \text{ by } (1.239)}, \tag{1.240}$$

so, using (VE) on the left and (CB) on the right,

$$\beta_*\|u-u_h\|_V^2 \le B(u-u_h,u-u_h) = B(u-u_h,u-v) \le \beta^*\|u-u_h\|_V\|u-v\|_V, \tag{1.241}$$

and cancelling one factor $\|u-u_h\|_V$,

$$\boxed{\|u-u_h\|_V \le \frac{\beta^*}{\beta_*}\|u-v\|_V} \qquad \square \tag{1.242}$$

Note (error estimate in $\|\cdot\|_V$). Take $v = \mathcal{I}_h u$, the projection of $u \in V$ into V_h. Then

$$\|u-u_h\|_V \le \frac{\beta^*}{\beta_*}\|u-\mathcal{I}_h u\|_V, \tag{1.243}$$

where $\|u-\mathcal{I}_h u\|_V$ is the *interpolation* error measured in $\|\cdot\|_V$. If you can generate a sequence of meshes and associated bases such that $\|u-\mathcal{I}_h u\|_V \to 0$ as $h \to 0$, then you have convergence; typically

$$\|u-\mathcal{I}_h u\|_V = O(h^p). \tag{1.244}$$

Example: the 1D Dirichlet problem, checked against the three conditions

$$-u'' = f \ \text{ in } (0,1), \qquad u(0)=u(1)=0, \qquad I=(0,1), \qquad V = H_0^1(I), \tag{1.245}$$

$$\|v\|_V^2 = \int_0^1 \left[(v')^2 + v^2\right] \mathrm{d}x = \|v'\|_{L^2(I)}^2 + \|v\|_{L^2(I)}^2, \tag{1.246}$$

$$B(u,v) = \int_0^1 u'v' \,\mathrm{d}x, \qquad L(v) = \int_0^1 fv \,\mathrm{d}x. \tag{1.247}$$

Check conditions.

1) **Symmetric:** $(u',v') = (v',u')$. ✓
2) **Continuous / bounded:** $|B(u,v)| \le \|u'\|_{L^2(I)}\|v'\|_{L^2(I)} \le \|u\|_V\|v\|_V$, so $\beta^* = 1$. ✓
3) **V-elliptic:** $B(v,v) = \|v'\|_{L^2(I)}^2 \ge \frac{1}{2}\left(\|v'\|_{L^2(I)}^2 + \|v\|_{L^2(I)}^2\right) = \frac{1}{2}\|v\|_V^2$, using **Poincaré** $\|v\|_{L^2(I)} \le \|v'\|_{L^2(I)}$ for $v \in H_0^1(I)$; so $\beta_* = \frac{1}{2}$. ✓
4) **L continuous:** $|L(v)| \le \|f\|_{L^2(I)}\|v\|_{L^2(I)} \le \|f\|_{L^2(I)}\|v\|_V$, so $\widetilde{L} = \|f\|_{L^2(I)}$. ✓

$\Rightarrow$ existence, uniqueness and boundedness, and Céa's lemma gives $\|u-u_h\|_V \le 2\|u-v\|_V$ for all $v \in V_h$, hence

$$\text{CG(1)}: \ \|u-u_h\|_V \le c_1\|h\,u''\|_{L^2(I)}, \qquad \text{CG(2)}: \ \|u-u_h\|_V \le c_2\|h^2 u'''\|_{L^2(I)}. \tag{1.248}$$

Part III — two dimensions

The 2D Poisson equation

$$-\Delta u = -u_{,xx} - u_{,yy} = f \ \text{ on } \Omega \subset \mathbb{R}^2, \qquad u = 0 \text{ on } \partial\Omega. \tag{1.249}$$

Multiply by a test function v and integrate over Ω:

$$-\iint_\Omega v\,\Delta u \,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega fv \,\mathrm{d}x\,\mathrm{d}y. \tag{1.250}$$

Note (product rule). $\nabla\cdot(v\nabla u) = \nabla v\cdot\nabla u + v\,\Delta u$, so

$$-\iint_\Omega v\,\Delta u\,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega \nabla u\cdot\nabla v\,\mathrm{d}x\,\mathrm{d}y - \iint_\Omega \nabla\cdot(v\nabla u)\,\mathrm{d}x\,\mathrm{d}y. \tag{1.251}$$

By the **divergence theorem**,

$$\iint_\Omega \nabla\cdot(v\nabla u)\,\mathrm{d}x\,\mathrm{d}y = \oint_{\partial\Omega} v\,\nabla u\cdot\boldsymbol{n}\;\mathrm{d}s, \qquad \boldsymbol{n} = \text{outward unit normal to } \partial\Omega. \tag{1.252}$$

Assuming $v = 0$ on $\partial\Omega$, the boundary integral vanishes and

$$\iint_\Omega \nabla u\cdot\nabla v\,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f v\,\mathrm{d}x\,\mathrm{d}y. \tag{1.253}$$

So

$$V = H_0^1(\Omega) = \Big\{v:\ \iint_\Omega \big[v^2 + \|\nabla v\|^2\big]\,\mathrm{d}x\,\mathrm{d}y < \infty,\ \ v = 0 \text{ on } \partial\Omega\Big\}, \qquad \|\nabla v\|^2 = (v_{,x})^2 + (v_{,y})^2, \tag{1.254}$$

$$B(u,v) \coloneqq \iint_\Omega \nabla u\cdot\nabla v\,\mathrm{d}x\,\mathrm{d}y, \qquad L(v) \coloneqq \iint_\Omega f v\,\mathrm{d}x\,\mathrm{d}y. \tag{1.255}$$

Check conditions.

1) **Symmetric:** $B(u,v) = B(v,u)$. ✓
2) **Continuous / bounded:** $|B(u,v)| \le \|\nabla u\|_{L^2(\Omega)}\|\nabla v\|_{L^2(\Omega)} \le \|u\|_V\|v\|_V$, so $\beta^* = 1$. ✓
3) **V-elliptic:** $B(v,v) = \|\nabla v\|^2_{L^2(\Omega)} \ge \frac{1}{2}\big(\|\nabla v\|^2_{L^2(\Omega)} + \|v\|^2_{L^2(\Omega)}\big)$, using the **multi-dimensional Poincaré inequality**

$$\|v\|_{L^2(\Omega)} \le C\,|v|_{H^1(\Omega)} = C\|\nabla v\|_{L^2(\Omega)} \qquad \forall\, v \in H_0^1(\Omega), \tag{1.256}$$

giving $B(v,v) \ge \frac{1}{2}\min\big(\frac{1}{C^2},1\big)\|v\|_V^2$, i.e. $\beta_* = \frac{1}{2}\min\big(\frac{1}{C^2},1\big)$, a constant depending only on $\partial\Omega$. ✓
4) **L continuous:** $|L(v)| \le \|f\|_{L^2(\Omega)}\|v\|_{L^2(\Omega)} \le \|f\|_{L^2(\Omega)}\|v\|_V$, so $\widetilde{L} = \|f\|_{L^2(\Omega)}$. ✓

CG(1) in 2D: the triangulation

Mesh $\mathcal{T}_h$: subdivide Ω into a finite number of triangles K_j, $j = 1,2,\dots,M$, with

$$h_j \coloneqq \text{longest edge of } K_j, \tag{1.257}$$

$$h \coloneqq \max_{1\le j\le M} h_j, \tag{1.258}$$

$$\rho_j \coloneqq \text{diameter of the largest circle inscribed in } K_j. \tag{1.259}$$

Defn. $\mathcal{T}_h$ is a **regular** triangulation if there exists a constant $\alpha > 0$, *independent of* h, such that

$$\frac{h_j}{\rho_j} \le \alpha \qquad \forall\, j. \tag{1.260}$$

The discrete space is

$$V_h = \Big\{v \in H^1(\Omega):\ v\big|_{K_j} = a_j + b_j x + c_j y,\ \ v\big|_{\partial\Omega} = 0\Big\}, \tag{1.261}$$

and Céa's lemma gives the error estimate

$$\|u - u_h\|_{H^1(\Omega)} \le c\,\|h\,u\|_{H^2(\Omega)}, \qquad |v|^2_{H^2(\Omega)} = \iint_\Omega \big[(v_{,xx})^2 + 2(v_{,xy})^2 + (v_{,yy})^2\big]\,\mathrm{d}x\,\mathrm{d}y. \tag{1.262}$$

The beam, in the abstract language

The 1D fourth-order problem written as an abstract variational problem:

$$u^{(iv)} = f \ \text{ in } (0,1), \qquad u(0) = u(1) = u'(0) = u'(1) = 0, \qquad V = H_0^2(I), \ \ I = (0,1), \tag{1.263}$$

$$\|v\|_V = \|v\|_{H^2(\Omega)[I]} = \left(\int_0^1 \left[v^2 + (v')^2 + (v'')^2 \right] \mathrm{d}x \right)^{1/2}, \tag{1.264}$$

$$B(u,v) = (u'', v''), \qquad L(v) = (f, v). \tag{1.265}$$

Here $H_0^2(I)$ means $u = 0$ *and* $u' = 0$ on the boundary.
Check conditions.

1) **Symmetric:** $(u'', v'') = (v'', u'')$. ✓
2) **Continuous / bounded:** $|B(u,v)| \le \|u''\|_{L^2(I)} \|v''\|_{L^2(I)} \le \|u\|_V \|v\|_V$, $\beta^* = 1$. ✓
3) V**-elliptic:** $B(v,v) = \|v''\|_{L^2(I)}^2$, and applying Poincaré twice, $\|v\|_{L^2(I)} \le \|v'\|_{L^2(I)} \le \|v''\|_{L^2(I)}$ for $v \in H_0^2(I)$, gives $B(v,v) \ge \frac{1}{3}\left(\|v\|_{L^2}^2 + \|v'\|_{L^2}^2 + \|v''\|_{L^2}^2\right) = \frac{1}{3}\|v\|_V^2$, so $\beta_* = \frac{1}{3}$. ✓
4) L **continuous:** $\widetilde{L} = \|f\|_{L^2(0,1)}$. ✓

CG(3) estimate (Céa's lemma):

$$\|u - u_h\|_{H^2(\Omega)[(0,1)]} \le c \, \|h^2 u\|_{H^4(\Omega)[(0,1)]}, \qquad |v|_{H^4(\Omega)[(0,1)]}^2 = \int_0^1 \left(v''''\right)^2 \mathrm{d}x. \tag{1.266}$$

Recall (this is HW3 Problems 4–5, done abstractly). The homework proved $\|u - U\|_E = O(h^2)$ by hand, with Galerkin orthogonality, Cauchy and the interpolation estimate. (1.266) is the same result read straight off Céa's lemma with $\beta^*/\beta_* = 3$. The hand proof gave a better constant because it used the energy norm, in which $\beta^* = \beta_*$; the abstract route gives the same *rate* with less work.

The biharmonic equation [12/Oct/26]

The 2D plate — the beam's two-dimensional counterpart:

$$\Delta^2 u = \Delta(\Delta u) = f \implies u_{,xxxx} + 2u_{,xxyy} + u_{,yyyy} = f, \tag{1.267}$$

$$\text{BCs:} \qquad u\big|_{\partial\Omega} = 0, \qquad \nabla u \cdot \boldsymbol{n}\big|_{\partial\Omega} = 0. \tag{1.268}$$

Multiply by v and integrate, letting $z = \Delta u$:

$$\begin{aligned} \iint_\Omega v \, \Delta(\Delta u) \, \mathrm{d}x \, \mathrm{d}y &= \iint_\Omega v f \, \mathrm{d}x \, \mathrm{d}y, \\ \iint_\Omega v \, \Delta z \, \mathrm{d}x \, \mathrm{d}y &= -\iint_\Omega \nabla v \cdot \nabla z \, \mathrm{d}x \, \mathrm{d}y + \oint_{\partial\Omega} v \, \nabla z \cdot \boldsymbol{n} \, \mathrm{d}s, \end{aligned} \tag{1.269}$$

and taking $v = 0$, $\nabla v \cdot \boldsymbol{n} = 0$ on $\partial\Omega$,

$$B(u,v) = \iint_\Omega \Delta u \, \Delta v \, \mathrm{d}x \, \mathrm{d}y, \qquad L(v) = \iint_\Omega f v \, \mathrm{d}x \, \mathrm{d}y, \qquad V = H_0^2(\Omega). \tag{1.270}$$

Note. If $v \in H_0^2(\Omega)$ then

$$\|v\|_{H^2(\Omega)}^2 = \|v\|_{L^2(\Omega)}^2 + \|\nabla v\|_{L^2(\Omega)}^2 + \|v_{,xx}\|_{L^2(\Omega)}^2 + 2\|v_{,xy}\|_{L^2(\Omega)}^2 + \|v_{,yy}\|_{L^2(\Omega)}^2, \tag{1.271}$$

and $v \in H_0^2(\Omega)$ means $v = 0$, $v_{,x} = 0$, $v_{,y} = 0$ on $\partial\Omega$.
Check conditions.

1) **Symmetric:** $B(u,v) = B(v,u)$. ✓

2) **Continuous / bounded:** $|B(u,v)| \le \|\Delta u\|_{L^2(\Omega)}\|\Delta v\|_{L^2(\Omega)} \le \|u\|_V\|v\|_V$. ✓

3) V**-elliptic:** $B(v,v) = \|\Delta v\|^2_{L^2(\Omega)}$, and by a result of **Girault–Raviart (1981)**,

$$\|v\|_{H^2(\Omega)} \le C\,\|\Delta v\|_{L^2(\Omega)} \qquad \text{for } v \in H^2_0(\Omega), \tag{1.272}$$

hence $B(v,v) \ge \frac{1}{C^2}\|v\|^2_V$. ✓

4) L **continuous:** $\widetilde{L} = \|f\|_{L^2(\Omega)}$. ✓

Conforming FEM for the biharmonic: the Argyris element

A conforming method needs $V_h \subset V = H^2_0(\Omega)$, i.e.

$$V_h = \left\{ v \in H^2(\Omega) :\ v\big|_{K_j} = \sum_{\ell+m=0}^{\ell+m\le 5} a_j^{(\ell m)}\, x^{(\ell)} y^{(m)},\ \ v\big|_{\partial\Omega} = 0,\ \ \nabla v\cdot \boldsymbol{n}\big|_{\partial\Omega} = 0 \right\}, \tag{1.273}$$

a complete quintic in two variables — **21 different polynomials** on each triangle. The corresponding element is the **Argyris element (1968)**, whose 21 degrees of freedom per triangle are

$$\underbrace{u,\ \nabla u,\ u_{,xx},\ u_{,xy},\ u_{,yy}}_{\text{6 per vertex} \times \text{3 vertices} = 18} \qquad \text{plus} \qquad \underbrace{\nabla u\cdot \boldsymbol{n}}_{\text{1 per edge} \times \text{3 edges} = 3}\ . \tag{1.274}$$

Part IV — 2D mesh generation

Triangulations

Note. FEM has no inherent restriction on the mesh. The simplest and most flexible choice is to use a **simplex**: 2D triangles, 3D tetrahedra.

Note. A general triangulation in 2D is **unstructured**, meaning grid points do not lie along coordinate lines.

Defn (triangulation). Let Ω be a 2D domain with polygonal boundary $\partial\Omega$. A triangulation $\mathcal{T}_h = \{K_j\}_{j=1}^M$ is a subdivision of Ω into M non-overlapping triangles constructed so that no vertex of one triangle lies on the edge of another triangle:

$$\bigcup_{j=1}^{M} K_j = \Omega. \tag{1.275}$$

Let $X_h = \{x_i = (x_i, y_i)\}_{i=1}^N$ denote the set of **nodes** (vertices of the triangles) and $\mathcal{S}_h = \{s_k\}_{k=1}^E$ the **edges**.

Main question (Mesh Generation, Part 1). Given a collection of points in $\mathbb{R}^2$ (nodes), how can we connect them to form a triangulation?

Voronoi diagrams

Idea. Given a finite set of points in $\mathbb{R}^2$, assign to each point a *region of influence* in such a way that it decomposes $\mathbb{R}^2$.

Defn (Voronoi region). Let X be a collection of N points in $\mathbb{R}^2$. For each $x_i \in X$ the **Voronoi region** V_i is the set of points $x \in \mathbb{R}^2$ that are at least as close to x_i as to any other point in X:

$$V_i = \left\{x \in \mathbb{R}^2 : \ \|x - x_i\| \leq \|x - x_j\| \ \ \forall\, x_j \in X\right\}. \tag{1.276}$$

Consider the half-plane

$$H_{ij} = \left\{x \in \mathbb{R}^2 : \ \|x - x_i\| \leq \|x - x_j\|\right\}. \tag{1.277}$$

The Voronoi region of x_i is the convex intersection of the half-planes H_{ij} for all $j \neq i$; therefore V_i is a convex polygonal region (possibly unbounded).

Notes.

(1) Each point in $\mathbb{R}^2$ has at least one nearest point in X, so every $x \in \mathbb{R}^2$ lies in at least one Voronoi region.

(2) Two Voronoi regions lie on opposite sides of a perpendicular bisector; two Voronoi regions do not share any interior points. If a point belongs to more than one region, it must lie on the boundary.

(3) Voronoi regions for a given X cover $\mathbb{R}^2$.

(4) Voronoi regions together with edges and vertices form the **Voronoi diagram** of X.

(5) For a given X, the Voronoi diagram is unique.

Delaunay triangulation

Defn (Delaunay partition). Let $X \subset \mathbb{R}^2$ be a set of N points and let V be the Voronoi diagram. The **Delaunay partition** is obtained by creating an edge between two points in X *if and only if* their Voronoi regions share a boundary.

Generic situation: 3 points of X share a common vertex, so a triangle is formed.

Degenerate situation: 4 or more points share a common Voronoi vertex — i.e. they are co-circular.

Note. Degeneracies can always be converted to a triangulation (a square is split by either diagonal).

Defn (Delaunay triangulation). The Delaunay triangulation $\mathcal{T}_h$ is the Delaunay partition together with some strategy to convert degenerate elements into triangles.

Properties.

(1) The Voronoi diagram and the Delaunay triangulation form a **duality**.

(2) For a given $X \subset \mathbb{R}^2$, V is unique, and $\mathcal{T}_h$ is unique up to degeneracies.

(3) $\mathcal{T}_h$ contains $O(N)$ triangles.

(4) $\mathcal{T}_h$ **maximises the minimum angle**.

(5) A circle circumscribing any triangle in $\mathcal{T}_h$ does not contain any other point of X.

In Python

1) Create a list of points in $\mathbb{R}^2$: `p[0:N, 0:2]`, where the first index is the point index and the second the coordinate (x, y).

2) Create $\mathcal{T}_h$ (Delaunay triangulation):

```
import scipy.spatial as sp
t = sp.Delaunay(p).simplices
```

3) Plot the mesh:

```
import numpy as np
import matplotlib.pyplot as plt
plt.triplot(p[:,0], p[:,1], t, facecolors=np.zeros(N), edgecolors='k')
```

4) Plot a function defined on the nodes:

```
U = ufunc(p)
plt.tripcolor(p[:,0], p[:,1], t, U, shading='flat', cmap='inferno')
```

What is `t`? N = number of nodes, M = number of triangles; `t` is an $M \times 3$ array of integers, where `t[j][i]` gives, for triangle j, the node index $i \in \{0, 1, 2\}$.

Note (area of a triangle). For vertices (x_0, y_0), (x_1, y_1), (x_2, y_2), set $\underline{v}_1 = (x_1 - x_0,\ y_1 - y_0,\ 0)$ and $\underline{v}_2 = (x_2 - x_0,\ y_2 - y_0,\ 0)$; then

$$\text{area} = \tfrac{1}{2}\left\|\underline{v}_1 \times \underline{v}_2\right\|. \tag{1.278}$$

Note. Delaunay in C: `http://www.qhull.org`.

Mesh generation, Part 2: representing Ω and $\partial\Omega$

Idea. Describe the domain *implicitly* using a **level-set function**.

Ex. $\phi(x) = \|x\| - R$, where $R > 0$. Then

$$\partial\Omega = \left\{x \in \mathbb{R}^2 :\ \phi(x) = 0\right\} = \left\{x \in \mathbb{R}^2 :\ \|x\| = R\right\}, \qquad \Omega = \left\{x \in \mathbb{R}^2 :\ \phi(x) < 0\right\}, \tag{1.279}$$

a circle of radius R centred at $(0, 0)$.

Note (convention).

$$\{\phi = 0\} \text{ is the boundary}, \qquad \{\phi < 0\} \text{ is the interior of } \Omega, \qquad \{\phi > 0\} \text{ is the exterior of } \Omega. \tag{1.280}$$

Note. For a given $\partial\Omega$ there are many level-set functions; e.g. $\phi(x) = \|x\|^p - R^p$ for any positive integer p describes the disc of radius R centred at $(0, 0)$.

Boolean operations.

$$\phi = \min(\phi_1, \phi_2) \qquad \text{is the } \textbf{union} \text{ operation: } \Omega = \Omega_1 \cup \Omega_2, \tag{1.281}$$

$$\phi = \max(\phi_1, \phi_2) \qquad \text{is the } \textbf{intersection} \text{ operation: } \Omega = \Omega_1 \cap \Omega_2. \tag{1.282}$$